Coefficient choice in Galois-ring lifts of APN permutations: sharp unit-direction spectra and one-term optimal lifts

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Abstract

Permutation polynomials over the Galois ring $\mathrm{GR}(2^k, m)$ are nonlinear components for symmetric primitives that add keys in the ring, as in secure computation modulo 2^k. Rønjom and Sandrib showed that their differential uniformity is at least $2q^{k-1}$, where $q = 2^m$. For the class studied here it is attained on top-layer differences, while almost all differences are units. We ask how small the difference table can be on unit differences when the residue permutation is fixed and only higher coefficient digits of its lift are chosen. Let $m \geq 5$ be odd, $k \geq 3$, and let p be a reduced quadratic almost perfect nonlinear (APN) permutation of $\mathbb{F}_q$ whose derivative is nowhere zero and two-to-one. Over all lifts of p, the least possible maximum on unit differences is q or $2q$, decided by two invariants of p and the level k. We describe every lift attaining it and show that the coefficients modulo 8 decide optimality at every level. For every odd $m \geq 5$, residue permutations with at most 16 monomials attain q after one 2-adic digit of one coefficient of the natural lift is changed. For an explicit infinite family, one added monomial lowers the maximum on unit differences from $4q$, or from at least q^2, to $2q$ at every level without raising the degree. Under a uniform starting state and independent uniform round keys, valuation preservation yields layerwise bounds for characteristics and two-round differentials that distinguish lifts with the same global uniformity. Core results are formalized in Lean 4.

Keywords: Galois rings; APN permutations; differential uniformity; coefficient lifts; differential characteristics; formal verification.
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1 Introduction

Let $R_k = \mathrm{GR}(2^k, m)$ be the Galois ring of characteristic 2^k whose residue field is $\mathbb{F}_q$, $q = 2^m$. Arithmetic in $\mathbb{Z}/2^k$ and in Galois rings over it is the native arithmetic of several protocols for secure multiparty computation [6, 7]. In such settings additions are cheap and nonlinear operations are ring multiplications, and over prime fields the same cost model has led to symmetric primitives built from low-degree polynomials [8]. A primitive over R_k in this spirit adds round keys in R_k and uses a permutation polynomial of R_k as its nonlinear layer. Difference propagation through this nonlinear layer is described by its difference table with respect to addition in R_k. The largest nontrivial entry is its differential uniformity in the sense of Nyberg [15], a basic parameter in differential cryptanalysis [14].

Rønjom and Sandrib [1] proved that every permutation polynomial of R_k has differential uniformity at least $2q^{k-1}$, and concluded that polynomial nonlinear layers over Galois rings are far

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from optimal. For the permutations considered below, their lower bound is attained on the top layer $2^{k-1}R_k$ of input differences, which contains a fraction $(q-1)/(q^k-1)$ of the nonzero differences. Almost all differences are units, and every lift considered here keeps the 2-adic valuation of a difference (Proposition 9.1). The behaviour on unit differences is therefore a separate design quantity, and so is the behaviour on each intermediate layer. For lifts whose residue is APN, the general upper bound on unit differences is $2q^{k-1}$ [1, Prop. 4(1)].

This paper determines how far the choice of coefficients can improve the unit differences. We fix a reduced quadratic APN permutation p of $\mathbb{F}_q$ satisfying hypothesis (H) below, where reduced means degree less than q, and consider its lifts: the polynomials over R_k that reduce to p coefficient by coefficient. We determine the least possible largest entry of the difference table on unit differences, all lifts attaining it, and the cost of attaining it. All lifts induce the same permutation of $\mathbb{F}_q$ and have the same differential uniformity $2q^{k-1}$ (for $\alpha \neq 0$ the solutions of $F(x+\alpha)-F(x)=\beta$ lie in at most two residue classes [1, 4]). On unit differences they behave very differently: depending on the lift, the largest entry ranges from the minimum (q or $2q$) to q^2 and more.

Terms. A *lift* of p is a polynomial $F \in R_k[X]$ whose coefficients reduce to those of p; the *natural lift* replaces each coefficient by the element of R_k with the same binary coordinates in the polynomial basis $1, t, \ldots, t^{m-1}$ (Section 3). We call k the *level*. We write $\delta_F(\alpha,\beta)$ for the number of $x \in R_k$ with $F(x+\alpha)-F(x)=\beta$. We assume that p' is nowhere zero and two-to-one, with $p'(a)=p'(b)$ exactly when $b \in \{a, a+\rho\}$ for a fixed $\rho \neq 0$ (hypothesis (H), Section 3). A unit difference α is *exceptional* if its residue is ρ and *ordinary* otherwise. Every lift has entries at most 2 in ordinary directions (Lemma 3.3), so everything happens in the exceptional directions, and we write $\Delta_{\mathrm{exc}}(F)$ for the largest entry there. Differences in $2^s R_k \setminus 2^{s+1}R_k$ form the *layer* s; the unit differences are layer 0 and the top layer is $s=k-1$.

Results. The minimum of Δ_{exc} over all lifts, of any degree, is q or $2q$ (Theorem B). Which value occurs depends on whether $k=3$ or $k \geq 4$ and on two invariants of p: the quadratic coefficient μ of the difference $p(X+\rho)+p(X)$, and a zero set $Z \subseteq \mathbb{F}_q$ (both defined in Section 2). Theorem C characterizes every lift attaining the minimum. Its criterion uses a function A_F on $\mathbb{F}_q$, which decides whether the solutions over a residue class split into several classes, and, when $\mu \neq 0$, a bit that decides whether the solutions coming from the two residues a and $a+\rho$ overlap. Optimality can be read off from the coefficients modulo 8 (Theorem D, Corollary E): a minimizer modulo 8 remains optimal at every higher level exactly when A_F has no zero, and a family of lifts that is optimal at every level exists exactly when $\mu=0$ or $Z \neq \emptyset$. The minimum is also cheap to reach. For the infinite family $P_{m,r}$ of quadratic APN permutations, one added monomial gives an optimal lift at every level without raising the degree, and for $r \geq 5$ a change of one coefficient digit does the same (Theorem A, Proposition 10.3). For every odd $m \geq 5$ there are residue permutations with at most 16 monomials for which the minimum is q, and a suitable change of one 2-adic digit of one coefficient of the natural lift attains it (Theorems B and 10.2, Proposition 10.3).

For example, let $r=3$ and $3 \nmid m$. The natural lift F of $P_{m,3}$ has $\Delta_{\mathrm{exc}}(F)=4q$. The lift $F+2X^{4+2^{m-1}}$ has $\Delta_{\mathrm{exc}}=2q$, and no lift of $P_{m,3}$ does better. A unit input difference then propagates with probability at most $2/q^{k-1}$, which is q^{k-2} times smaller than the general unit bound.

The proofs are local. Along a residue class the difference equation reduces to the quadratic map $y \mapsto \mu y^2 + A_F(a)y$ over $\mathbb{F}_q$, and one second-order invariant records how the solutions over a and $a+\rho$ are shifted against each other. The same analysis gives the full spectrum of every exceptional row when A_F has no zero, an exact formula for the penultimate layer, and the intermediate layers.

It also shows a conflict: when $\mu \neq 0$, optimizing the middle layer at $k = 3$ forces the exceptional directions at every $k \geq 4$ far from the optimum.

Cryptographic consequences. The choice of coefficients becomes a design parameter with a sharp target and a certificate that can be checked modulo 8. Since lifts preserve valuations, the bound for each layer combines with the branch number B of the reduced linear layer as in the wide-trail argument [10, 11], under a uniform starting state and independent uniform round keys added in R_k (Corollary G, Proposition 9.4). The top layer sets the bound over all inputs: for every lift, every differential over two or more rounds has probability at most $(2/q)^{B-1}$, and the bound for characteristics with input in the top layer is $(2/q)^{B\lfloor N/2 \rfloor}$. The coefficient choice improves the bounds for inputs below the top layer. In an eight-round example with four 21-bit words and a linear layer of branch number 5, the bound for characteristics with a unit input word improves from 2^{-140} for the natural lift to 2^{-260} for the lift $F^\sharp$ of $P_{7,5}$ with one added monomial, while the top layer gives 2^{-120} for every lift. The same valuation argument bounds two-round differentials, without the factor up to 2^n that a count over the active words would introduce for n words: a differential whose input has a unit word has probability at most 2^{-52} for $F^\sharp$ against 2^{-28} for the natural lift, and the bound over all differentials is 2^{-24} for both (Propositions 9.5 and 9.6). Markov's inequality then bounds the proportion of keys for which a fixed differential is large (Proposition 9.7). A fixed plaintext breaks the product formula (Example 9.2), so the probability model matters.

These results concern the polynomial part of a design. A cipher built only from ring polynomials, additive linear layers and keys added in R_k maps inputs that are congruent modulo 2 to outputs that are congruent modulo 2 (Proposition 9.8), so a complete design contains a further component that breaks this partition. Our theorems certify the polynomial segments and the coefficients chosen inside them.

Relation to other work. This paper is a substantially revised and extended version of the author's preprint [5], from which the family $P_{m,r}$ and its analysis on ordinary directions and on the top layer are taken. The minimax theorem, the classification of minimizers, the spectra and their certification modulo 8, the results across levels and layers, the optimal lifts of sparse representatives and the differential bounds are new. The earlier paper [4] constructs optimal reduced representatives of quadratic APN permutations in odd dimension; here the representative is fixed and the coefficients of its lift are optimized. A separate manuscript of the author, currently under review, leaves the polynomial setting and studies digit-exchanged permutations of $\mathrm{GR}(4, m)$; it shares the lifting framework and the basic directional estimates with the present paper. Bartoli and Stănică [3] study when reduced polynomial lifts of APN permutations are permutations. They give infinite liftable families, including a trace family of degree $q/2 + 2$ with a two-to-one derivative [3, Thm. 2.4], and a liftability criterion for reduced Gold-power representatives [3, Thm. 6.1, Cor. 6.2]. The residue polynomials $P_{m,r}$ fall under their Corollary 6.2, and $P_{m,3}$ is their Remark 6.3. They do not consider the difference tables of individual lifts. Their trace family satisfies (H), and we apply our theorems to it in Section 10. The dependence of the difference table on the lift is easy to see: over $\mathbb{Z}/8$ the permutations X and $X + 2X^2$ have the same residue but different differences in direction 1. What is new here is the sharp minimum for a fixed residue, its minimizers, and their certification and construction.

Verification. The central results are formalized in Lean 4, and the finite computations behind the examples and the existence proof for the sparse family can be replayed from Online Resource 1. Appendix A lists what is formalized and what remains on paper.

Organization. Section 2 states the main results. Section 3 sets up notation and the local expansion along a residue class. Section 4 introduces the lift invariants and shows how to prescribe them. Section 5 proves the exceptional spectra and the lower bounds, and Section 6 the minimax theorem, the classification of minimizers and optimality across levels. Section 7 treats $P_{m,r}$, Section 8 the lower layers, and Section 9 the cryptographic consequences. Section 10 discusses residue representatives, the trace family of [3] and open problems. Appendix A describes the computations, the formalization and the use of AI tools.

2 Main results

Throughout, $m \geq 5$ is odd, $q = 2^m$, $k \geq 3$ and $R_k = \mathrm{GR}(2^k, m)$. We fix a reduced quadratic APN permutation $p \in \mathbb{F}_q[X]$ satisfying (H) of Section 3. In particular $p' = \varepsilon + K$ with K linearized, $\ker K = \{0, \rho\}$, $\rho \neq 0$ and $\varepsilon \notin K(\mathbb{F}_q)$. A *lift* of p is any $F \in R_k[X]$ whose coefficients reduce to those of p. We impose no bound on the degree of F. A direction $\alpha \in R_k$ is *exceptional* if $\overline{\alpha} = \rho$, and

$$\Delta_{\mathrm{exc}}(F) = \max\{\delta_F(\alpha, \beta) : \ \overline{\alpha} = \rho, \ \beta \in R_k\},$$
$$\delta_F(\alpha, \beta) = \#\{x \in R_k : F(x + \alpha) - F(x) = \beta\}.$$

Two invariants depend only on p:

$$\mu = [X^2]\big(p(X + \rho) + p(X)\big), \qquad Z = \{a \in \mathbb{F}_q : \ p(a + \rho) + p(a) + \rho\, p'(a) = 0\}.$$

Two invariants depend on the lift. For $\overline{x} = a$ and $\overline{\tau} = \rho$ put

$$A_F(a) = \overline{\tfrac{1}{2}\big(F'(x + \tau) - F'(x)\big)},$$
$$E_F(a) = \overline{\tfrac{1}{4}\big(F(x + 2\tau) - 2F(x + \tau) + F(x)\big)} \quad (a \in Z).$$

Both are well defined (Lemmas 4.1 and 4.3). When $\mu \neq 0$, $a \in Z$ and $A_F(a) \neq 0$ we set $\chi_F(a) = \mathrm{Tr}\big(\mu E_F(a)/A_F(a)^2\big)$.

Theorem A (A one-term optimal lift)**.** *Let $m \geq 5$ be odd, $3 \leq r < m$ odd with $\gcd(r, m) = 1$, and let F be the natural 0/1 lift of the explicit APN permutation $P_{m,r}$ of Section 7. Put*

$$F^\sharp = F + 2X^3 \ (r \geq 5), \qquad F^\sharp = F + 2X^{4+2^{m-1}} \ (r = 3).$$

For every $k \geq 3$ every exceptional row of $F^\sharp$ consists of exactly $q^{k-1}/2$ entries equal to $2q$ and zeros, and every ordinary unit row has entries at most 2. Thus $\Delta_{\mathrm{unit}}(F^\sharp) = 2q$, which is the least value of Δ_{exc} over all lifts of $P_{m,r}$. The correction does not raise the degree $3q/4$ and, in the termwise evaluation model, costs one ring multiplication and two additions on top of the squaring schedule of F; for $r \geq 5$ a change of one coefficient digit gives the same spectra with no extra multiplication (Proposition 10.3). The natural lift itself has $\Delta_{\mathrm{exc}} = 4q$ for $r = 3$ and $\Delta_{\mathrm{exc}} \geq q^2$ for $r \geq 5$ at every level; at $k = 3$ and $r \geq 5$ its value is exactly q^2 if $r \equiv 1 \pmod 4$ and $2q^2$ if $r \equiv 3 \pmod 4$ (Corollary 7.5).

Theorem B (Minimax over all lifts)**.** *The smallest possible value of $\Delta_{\mathrm{exc}}(F)$ over all lifts F of p is*

	$k=3$	$k\geq 4$
$\mu=0,\ Z=\emptyset$	q	q
$\mu=0,\ Z\neq\emptyset$	$2q$	$2q$
$\mu\neq 0,\ Z=\emptyset$	q	$2q$
$\mu\neq 0,\ Z\neq\emptyset$	$2q$	$2q$

Theorem C (All minimizers)**.** *Let $k\geq 4$. If $\mu=0$, a lift is optimal if and only if A_F has no zero. If $\mu\neq 0$, a lift is optimal if and only if A_F has no zero and $\chi_F\equiv 1$ on Z. For $k=3$ the optimal lifts are those with A_F zero-free when $\mu=0$, those with $A_F\equiv 0$ when $\mu\neq 0$ and $Z=\emptyset$, and those with $\chi_F(a)=1$ at every $a\in Z$ with $A_F(a)\neq 0$ when $\mu\neq 0$ and $Z\neq\emptyset$.*

Theorem D (Exceptional spectra and certification modulo 8)**.** *Let n_c be the number of β with $\delta_F(\alpha,\beta)=c$ for a fixed exceptional α.*

(1) If A_F has no zero, then $\delta_F(\alpha,\beta)=\delta_{F\bmod 8}(\alpha \bmod 8,\beta \bmod 8)$. For $\mu=0$ we get $n_q=(q-|Z|)q^{k-2}$ and $n_{2q}=\frac{1}{2}|Z|q^{k-2}$. For $\mu\neq 0$ we get $n_{2q}=\frac{1}{2}(q-\nu)q^{k-2}$ and $n_{4q}=\frac{1}{4}\nu q^{k-2}$, where $\nu=\#\{a\in Z:\chi_F(a)=0\}$.

(2) If $k=3$ and $\mu\neq 0$, then for arbitrary A_F

$$n_q=(s-z_0)q,\quad n_{2q}=\tfrac{1}{2}(q-s+z_0-\nu)q,\quad n_{4q}=\tfrac{1}{4}\nu q,$$

with $s=\#A_F^{-1}(0)$, $z_0=\#(Z\cap A_F^{-1}(0))$ and $\nu=\#\{a\in Z:A_F(a)\neq 0,\ \chi_F(a)=0\}$.

All other positive values have multiplicity zero.

Corollary E (Optimality across levels)**.** *The criteria in Theorem C determine optimality at every higher level from the coefficient data modulo 8. A minimizer at level $k=3$ (that is, modulo 8) remains optimal at all higher levels if and only if its function A_F is nowhere zero. Consequently, a compatible family minimizing the exceptional uniformity at every level $k\geq 3$ exists if and only if $\mu=0$ or Z is nonempty.*

Theorem F (Penultimate layer and compatibility)**.** *Let $s=\#A_F^{-1}(0)$. For $k\geq 4$ and $\alpha\in 2^{k-2}R_k\setminus 2^{k-1}R_k$, every row has $q-s$ entries q^{k-1} and $s/2$ entries $2q^{k-1}$. Optimality of the $k=3$ middle layer $2R_3\setminus 4R_3$ (all nonzero entries equal to q^2) requires $A_F\equiv 0$ when $\mu\neq 0$, which forces $\Delta_{\mathrm{exc}}\geq q^2$ at every level $k\geq 4$. When $\mu=0$ a zero-free A_F optimizes all three at once.*

Corollary G (Transition probabilities)**.** *For $F^\sharp$ the maximal transition probability from a unit input difference is $2/q^{k-1}$, a factor q^{k-2} below the general bound of Rønjom and Sandrib. Differences keep their 2-adic valuation. Consider N rounds $x\mapsto M\cdot S(x)+\kappa_i$ on R_k^n, where S applies a lift F of p to each word and M is an additive automorphism of R_k^n whose reduction $\overline{M}$ has differential branch number B. Take the starting state and the round keys uniform and all independent, with the keys added in R_k.*

(a) If $F=F^\sharp$, a fixed characteristic whose input difference has a unit component has probability at most $(2/q^{k-1})^{B\lfloor N/2\rfloor}$.

(b) If $F=F^\sharp$ with $r\geq 5$ and the input difference lies in $(2^sR_k)^n$ with a word outside $2^{s+1}R_k$, $1\leq s\leq k-2$, the bound is $q^{(s+1-k)B\lfloor N/2\rfloor}$. For $s=k-1$ it is $(2/q)^{B\lfloor N/2\rfloor}$ for every lift.

(c) For every lift, every $N \geq 2$ and every pair (a, b) with $a \neq 0$, the differential $a \to b$ has probability at most $(2/q)^{B-1}$. If $F = F^\sharp$ and a has a unit component, it has probability at most $(2/q^{k-1})^{B-1}$; if $F = F^\sharp$ with $r \geq 5$ and a is in layer s as in (b), $1 \leq s \leq k-2$, at most $q^{(s+1-k)(B-1)}$.

(d) For each such two-round differential with bound ε from (c) and every $t > 0$, at most a proportion ε/t of the key pairs gives it a fixed-key probability $\geq t$.

3 Preliminaries

3.1 Rings, lifts and direction classes

Let $f \in \mathbb{F}_2[t]$ be irreducible of degree m and $\tilde{f}$ its 0/1 lift. Then $R_k = (\mathbb{Z}/2^k)[t]/(\tilde{f})$ and $\mathbb{F}_q = \mathbb{F}_2[t]/(f)$, with reduction $x \mapsto \overline{x}$. The *natural lift* of $p \in \mathbb{F}_q[X]$ replaces each coefficient by the 0/1 lift of its coordinates in the basis $1, t, \ldots, t^{m-1}$; it depends on f. A nonzero $\alpha \in R_k$ is *ordinary* if $\overline{\alpha} \notin \{0, \rho\}$, *exceptional* if $\overline{\alpha} = \rho$, and lies in the layer v_j if $\alpha \in 2^j R_k \setminus 2^{j+1} R_k$ $(1 \leq j \leq k-1)$. The layer v_{k-1} is the top layer and v_{k-2} the penultimate layer. We write $\Delta_{\mathrm{cls}}(F)$ for the maximum of $\delta_F(\alpha, \beta)$ over α in a class and all β, and $\Delta_{\mathrm{unit}} = \max(\Delta_{\mathrm{ord}}, \Delta_{\mathrm{exc}})$.

For $F \in R_k[X]$ we use Hasse derivatives $F^{[i]}$, defined by $F(x+h) = \sum_{i \geq 0} F^{[i]}(x) h^i$ exactly in R_k. Then $F^{[1]} = F'$ and $2F^{[2]} = F''$. For a function g on $\mathbb{F}_q$ we write $D_\rho g(a) = g(a+\rho) + g(a)$. Division by 2 or 4 is applied only after the numerator has been shown to lie in $2R_k$ or $4R_k$, and only its residue is used.

3.2 The hypothesis (H)

Definition 3.1. A polynomial $p \in \mathbb{F}_q[X]$ satisfies (H) if it is reduced ($\deg p < q$), quadratic (only Dembowski–Ostrom monomials $X^{2^i+2^j}$, linear monomials X^{2^i} and a constant), an APN permutation of $\mathbb{F}_q$, and its formal derivative is $p' = \varepsilon + K$ with K linearized, $\ker K = \{0, \rho\}$, $\rho \neq 0$ and $\varepsilon \notin K(\mathbb{F}_q)$.

For $m = 3$ no polynomial satisfies (H): an exhaustive search over the reduced quadratic APN permutations of $\mathbb{F}_8$ finds none with $\varepsilon \notin K(\mathbb{F}_8)$ and $\#\ker K = 2$. This is why we assume $m \geq 5$. Under (H), p' has no zero and is exactly two-to-one: $p'(a) = p'(b)$ if and only if $b \in \{a, a+\rho\}$. Every lift of p is a permutation of R_k, because its residue is a permutation and its derivative takes unit values [1, Thm. 2], [3, Thm. 2.1]. Put $\Lambda(a) = p(a+\rho) + p(a) + \rho p'(a)$, so that $Z = \Lambda^{-1}(0)$, and $z = |Z|$.

Lemma 3.2. *(i) $p^{[2]}$ is affine, so $p^{[2]}(a+\rho) + p^{[2]}(a) = \mu$ for all a.*

(ii) Λ is ρ-periodic, so Z is a union of pairs $\{a, a+\rho\}$.

(iii) $a \mapsto p(a+\rho) + p(a)$ is ρ-periodic and exactly two-to-one.

Proof. (i) The second Hasse coefficient of X^e is $\binom{e}{2}$. For $e = 2^i + 2^j$ with $i < j$ it is odd only for $e = 3$ or $i = 1$, and for $e = 2^i$ only for $e = 2$. Hence $p^{[2]} = c_2 + c_3 X + \sum_{j \geq 2} c_{2+2^j} X^{2^j}$ is affine. (ii) $\Lambda(a+\rho) = p(a) + p(a+\rho) + \rho p'(a+\rho) = \Lambda(a)$ because $p'(a+\rho) = p'(a)$. (iii) Periodicity is clear. Since p is APN every value has at most two preimages, and periodicity gives exactly the two preimages a and $a+\rho$. □

Lemma 3.3 (Ordinary directions)**.** *If α is ordinary, then $\delta_F(\alpha, \beta) \in \{0, 2\}$.*

Proof. The residue $\overline{G} = p(X+\overline{\alpha}) + p(X) + \overline{\beta}$ of $G = F(X+\alpha) - F(X) - \beta$ has at most two roots by the APN property. At a root $\overline{G}'(a) = p'(a+\overline{\alpha}) + p'(a) = K(\overline{\alpha}) \neq 0$. Simple roots lift uniquely by Hensel's lemma. □

3.3 The local expansion along a residue class

Fix an exceptional α and $\beta \in R_k$ and put $G(X) = F(X+\alpha) - F(X) - \beta$.

Lemma 3.4 (The branch gate)**.** *Let $\overline{x} = a$.*

(i) $G(x+2u) \equiv G(x) \pmod 4$ *for all* $u \in R_k$.

(ii) $G(x+\alpha) \equiv G(x) + 2[\Lambda(a)] \pmod 4$, *where* $[\cdot]$ *is any lift.*

Hence, if $G \in 4R_k$ on one residue class of a pair, it lies in $4R_k$ on the other class if and only if $\Lambda(a) = 0$.

Proof. (i) $G(x+2u) = G(x) + 2uG'(x) + 4(\cdots)$ and $G'(x) = F'(x+\alpha) - F'(x) \in 2R_k$, because its residue is $p'(a+\rho) + p'(a) = 0$. (ii) $G(x+\alpha) - G(x) = S(x) := F(x+2\alpha) - 2F(x+\alpha) + F(x) = \sum_{i\geq 1}(2^i - 2)\alpha^i F^{[i]}(x)$. The term $i = 1$ vanishes and $2^i \equiv 0 \pmod 4$ for $i \geq 2$, so $S(x) \equiv -2\big(F(x+\alpha) - F(x) - \alpha F'(x)\big) \pmod 4$. The bracket has residue $\Lambda(a)$. □

Lemma 3.5 (Local formula)**.** *Let x_a be a fixed lift of a with $G(x_a) \in 4R_k$, put $c_a = \overline{G(x_a)/4}$ and $L_a(y) = \mu y^2 + A_F(a)y$.*

(i) Choose $g_0, g_1 \in R_k$ with $G(x_a) = 4g_0$ and $G'(x_a) = 2g_1$ (possible by Lemma 4.1(ii)) and put $\Phi_a(z) = g_0 + g_1 z + G^{[2]}(x_a)z^2 + \sum_{i\geq 3} 2^{i-2}G^{[i]}(x_a)z^i$. Then $G(x_a + 2z) = 4\Phi_a(z)$, so $G(x_a + 2z) = 0$ if and only if $\Phi_a(z) \in 2^{k-2}R_k$. This condition depends only on z modulo 2^{k-2}, and Φ_a reduces modulo 2^{k-2} to a polynomial over $R_k/2^{k-2}R_k \cong R_{k-2}$ whose residue is $c_a + A_F(a)y + \mu y^2$ with $y = \overline{z}$. The class of a is $\{x_a + 2z\}$ with z running modulo 2^{k-1}.

(ii) For $k = 3$ the number of solutions in the class of a is $q \cdot \#\{y \in \mathbb{F}_q : L_a(y) = c_a\}$.

(iii) If $\Lambda(a) = 0$ and $x_{a+\rho} = x_a + \alpha$, then $c_{a+\rho} = c_a + E_F(a)$.

Proof. (i) The residue of $G'(x_a)/2$ is $A_F(a)$ by definition, and that of $G^{[2]}(x_a)$ is $p^{[2]}(a+\rho) + p^{[2]}(a) = \mu$ by Lemma 3.2(i). (ii) For $k = 3$ the value $G(x_a + 2z)/4$ modulo 2 depends on $\overline{z}$ only, and each $\overline{z}$ has q lifts z modulo 4. (iii) $G(x_a+\alpha) - G(x_a) = S(x_a) \in 4R_k$ when $\Lambda(a) = 0$, and $\overline{S(x_a)/4} = E_F(a)$. □

4 Lift invariants

Lemma 4.1 (A_F)**.** *(i)* $F'(x+2u) \equiv F'(x) \pmod 4$ *for all* x, u.

(ii) $F'(x+\tau) - F'(x) \in 2R_k$ *when* $\overline{\tau} = \rho$, *and* $A_F(a)$ *does not depend on the lifts* x, τ.

(iii) A_F *is* ρ*-periodic.*

Proof. (i) $F'(x+2u) = F'(x) + 2uF''(x) + \sum_{i\geq 2}(2u)^i (F')^{[i]}(x)$ and $F'' = 2F^{[2]}$. (ii) The residue of the difference is $p'(a+\rho) + p'(a) = 0$. Changing x to $x + 2u$ or τ to $\tau + 2v$ does not change either term modulo 4 by (i). (iii) $A_F(a) + A_F(a+\rho)$ is the residue of $(F'(x+2\tau) - F'(x))/2$, using $x + \tau$ as the lift of $a + \rho$. By (i) the numerator lies in $4R_k$. □

Proposition 4.2 (Prescribing A_F). *(i) For $H \in R_k[X]$, the lift $F + 2H$ has the same residue and $A_{F+2H} = A_F + D_\rho h$, where h is the residue function of H'.*

(ii) Every function $h : \mathbb{F}_q \to \mathbb{F}_q$ is the residue function of H' for some H.

(iii) $\{D_\rho h\}$ is the set of all ρ-periodic functions.
Hence A_F can be changed to any ρ-periodic function without changing the residue of F.

Proof. (i) The change of the numerator is $2(H'(x+\tau) - H'(x))$. (ii) Let $s = h^{1/2}$ pointwise, interpolate $s = \sum s_e X^e$ and put $H = \sum_e [s_e^2] X^{2e+1}$. Then $H' \equiv \sum_e s_e^2 X^{2e} = s^2 \pmod 2$. (iii) A ρ-periodic g is $D_\rho h$ for $h(a) = 0$, $h(a+\rho) = g(a)$ on a set of pair representatives. □

Lemma 4.3 (E_F). *Let $a \in Z$, $\overline{x} = a$, $\overline{\tau} = \rho$ and $S(x) = F(x+2\tau) - 2F(x+\tau) + F(x)$.*

(i) $S(x) \in 4R_k$.

(ii) $\overline{S(x)/4}$ does not depend on the lifts x and τ.

Proof. (i) As in Lemma 3.4(ii), $S(x) \equiv -2(F(x+\tau) - F(x) - \tau F'(x)) \pmod 4$ and the bracket has residue $\Lambda(a) = 0$. (ii) Changing x to $x + 2u$ adds $2uS'(x) + 4u^2 S^{[2]}(x) + \sum_{i \geq 3} (2u)^i S^{[i]}(x)$. By Lemma 4.1(i), $S'(x) \equiv 2(F'(x) - F'(x+\tau)) \equiv 0 \pmod 4$, and $S^{[2]}(x)$ has residue $p^{[2]}(a) + p^{[2]}(a) = 0$. So the change lies in $8R_k$. Changing τ to $\tau + 2v$ changes S modulo 8 by $4v(F'(x+2\tau) - F'(x+\tau))$, and $F'(x+2\tau) \equiv F'(x) \equiv F'(x+\tau) \pmod 2$. □

Proposition 4.4 (Translation identity). *For $a \in Z$, $E_F(a+\rho) + E_F(a) = \rho A_F(a) + \rho^2 \mu = L_a(\rho)$.*

Proof. Put $G(X) = F(X+\tau) - F(X)$, so $S(X) = G(X+\tau) - G(X)$. The sum telescopes exactly: $S(x+\tau) + S(x) = G(x+2\tau) - G(x)$. Modulo 8, $G(x+2\tau) - G(x) \equiv 2\tau G'(x) + 4\tau^2 G^{[2]}(x)$. Here $G'(x) \in 2R_k$ with $\overline{G'(x)/2} = A_F(a)$, and $\overline{G^{[2]}(x)} = \mu$. Both a and $a+\rho$ lie in Z, and $x+\tau$ is a lift of $a+\rho$ by Lemma 4.3(ii). □

Lemma 4.5 (Overlap of images). *Put $A = A_F(a)$. (i) If $\mu \neq 0$ and $A \neq 0$, then L_a is $\mathbb{F}_2$-linear and two-to-one with kernel $\{0, A/\mu\}$, and $E \in L_a(\mathbb{F}_q)$ if and only if $\mathrm{Tr}(\mu E/A^2) = 0$. (ii) If $\mu \neq 0$ and $A = 0$, then L_a is bijective. (iii) If $\mu = 0$ and $A \neq 0$, then L_a is bijective; if $\mu = A = 0$, then $L_a \equiv 0$. Moreover $\chi_F(a+\rho) = \chi_F(a)$ for $a \in Z$.*

Proof. With $y = (A/\mu)t$ we get $L_a(y) = (A^2/\mu)(t^2+t)$, and $\{t^2+t\} = \ker \mathrm{Tr}$ (both have $q/2$ elements and $\mathrm{Tr}(t^2+t) = 0$). For the periodicity, $A_F(a+\rho) = A_F(a)$ and $\mathrm{Tr}(\mu(E_F(a+\rho) + E_F(a))/A^2) = \mathrm{Tr}(s+s^2) = 0$ with $s = \mu\rho/A$ by Proposition 4.4. □

Proposition 4.6 (Prescribing E_F). *For $J \in R_k[X]$ with residue function j, the lift $F + 2J^2$ has the same residue and the same A_F, and $E_{F+2J^2}(a) = E_F(a) + (D_\rho j(a))^2$ for $a \in Z$. Hence, keeping A_F, the function E_F can be shifted on Z by any ρ-periodic function.*

Proof. Put $Q = J^2$. Since $Q' = 2JJ'$, we have $Q(x+2\tau) \equiv Q(x) \pmod 4$. The change of E is the residue of $(Q(x+2\tau) - 2Q(x+\tau) + Q(x))/2 \equiv Q(x) - Q(x+\tau) \pmod 2$, which is $j(a)^2 + j(a+\rho)^2$. The change of A is D_ρ of the residue of Q', which is 0. The last sentence follows from Proposition 4.2(iii) and bijectivity of squaring. □

5 Exceptional spectra and lower bounds

Theorem 5.1. *Let $k = 3$ and fix an exceptional α.*

(i) If $\mu \neq 0$, then $n_q = (s - z_0)q$, $n_{2q} = (q - s + z_0 - \nu)q/2$ and $n_{4q} = \nu q/4$.

(ii) If $\mu = 0$ and A_F has no zero, then $n_q = (q - z)q$ and $n_{2q} = (z/2)q$.

Proof. If $\overline{\beta}$ is not a value of $D_\rho p$ then $\delta_F(\alpha, \beta) = 0$. If $\overline{\beta} = D_\rho p(a)$ only the pair $\{a, a + \rho\}$ contributes (Lemma 3.2(iii)). By Lemma 3.4(i) the class of a passes the gate for exactly one value of β mod 4. The remaining digit of β runs through q values and is in bijection with c_a. The class of a contributes $q \cdot \#L_a^{-1}(c_a)$ (Lemma 3.5(ii)).

If $\Lambda(a) \neq 0$ the two classes pass for different $\beta \bmod 4$ (Lemma 3.4(ii)). For each class: if $\mu \neq 0 = A(a)$ we get q values β with entry q. If $\mu \neq 0 \neq A(a)$ we get $q/2$ values with entry $2q$. If $\mu = 0 \neq A(a)$ we get q values with entry q.

If $\Lambda(a) = 0$ both classes pass for the same $\beta \bmod 4$, and $c_{a+\rho} = c_a + E_F(a)$ (Lemma 3.5(iii)). The entry is $q(\#L_a^{-1}(c) + \#L_a^{-1}(c + E))$, since $L_{a+\rho} = L_a$ by Lemma 4.1(iii). If $\mu \neq 0 = A$ we get q values with entry $2q$. If $\mu \neq 0 \neq A$ and $\chi = 1$ the images are disjoint and we get q values with entry $2q$. If $\chi = 0$ the images coincide and we get $q/2$ values with entry $4q$. If $\mu = 0 \neq A$ we get q values with entry $2q$.

Counting residues outside and inside Z gives the formulas. For (i): $s - z_0$ residues outside Z with $A = 0$ give n_q. The value $2q$ comes from $q - z - s + z_0$ residues outside Z with $A \neq 0$ (each $q/2$), $z_0/2$ pairs in Z with $A = 0$, and $(z - z_0 - \nu)/2$ pairs in Z with $\chi = 1$ (each q). The value $4q$ comes from $\nu/2$ pairs with $\chi = 0$ (each $q/2$). As a check, $q\, n_q + 2q\, n_{2q} + 4q\, n_{4q} = q^3$. □

Theorem 5.2 (Stability modulo 8)**.** *If A_F has no zero, then $\delta_F(\alpha, \beta) = \delta_{F_3}(\alpha_3, \beta_3)$ for every exceptional α and every β, where the index 3 denotes reduction modulo 8. Consequently, for every $k \geq 3$, $n_q = (q - z)q^{k-2}$ and $n_{2q} = (z/2)q^{k-2}$ if $\mu = 0$, and $n_{2q} = (q - \nu)q^{k-2}/2$ and $n_{4q} = \nu q^{k-2}/4$ if $\mu \neq 0$.*

Proof. By Lemma 3.5(i), the solutions in the class of a correspond to the z modulo 2^{k-1} with $\Phi_a(z) \equiv 0 \pmod{2^{k-2}}$, and the residue $c_a + Ay + \mu y^2$ of Φ_a has constant derivative $A = A_F(a) \neq 0$. Every residue root is simple and lifts by Hensel's lemma over R_{k-2} to exactly one root modulo 2^{k-2}, which has q lifts modulo 2^{k-1}. So each residue root gives q solutions. The number of residue roots depends only on c_a (and on $c_a + E_F(a)$ when $\Lambda(a) = 0$), hence only on $\beta \bmod 8$. The same count holds at $k = 3$. Each $\beta \bmod 8$ has q^{k-3} lifts. The invariants A_F and E_F depend only on F modulo 8. □

Theorem 5.3 (Lower bounds)**.** *For every lift and every $k \geq 3$: (i) $\Delta_{\mathrm{exc}} \geq q$; (ii) $\Delta_{\mathrm{exc}} \geq 2q$ if $Z \neq \emptyset$; (iii) $\Delta_{\mathrm{exc}} \geq q^2$ if $k \geq 4$ and A_F has a zero; (iv) $\Delta_{\mathrm{exc}} \geq 2q$ if $k \geq 4$ and $\mu \neq 0$.*

Proof. (i) By Lemma 3.4(i) the q^{k-1} differences on a residue class lie in one coset of $4R_k$, which has q^{k-2} elements. (ii) If $\Lambda(a) = 0$ the two classes of the pair, $2q^{k-1}$ points, land in the same coset of $4R_k$. (iii) Let $A_F(a) = 0$ and $\overline{x} = a$. Then $G'_\alpha(x) \equiv 0 \pmod 4$. For $h \in 2^{k-2}R_k$, $G_\alpha(x + h) - G_\alpha(x) = hG'_\alpha(x) + \sum_{i \geq 2} h^i G_\alpha^{[i]}(x)$. The first term lies in $2^{k-2} \cdot 4R_k = 0$, and $h^i \in 2^{2k-4}R_k = 0$ for $k \geq 4$. So the difference is constant on the q^2 points of $x + 2^{k-2}R_k$. (iv) By (iii) we may assume A_F has no zero. Choose β with $c_a = 0$. Then $L_a(y) = 0$ has two roots, and by Theorem 5.2 each gives q solutions. □

6 Minimax, all minimizers and optimality across levels

Proof of Theorem B. The lower bounds are Theorem 5.3; for $k = 3$ and $\mu \neq 0$ only (i) and (ii) are used. For attainment: if $\mu = 0$, make $A_F \equiv 1$ (Proposition 4.2); Theorems 5.1 and 5.2 give the value $q(1 + [Z \neq \emptyset])$. If $\mu \neq 0$ and $k = 3$, make $A_F \equiv 0$; Theorem 5.1(i) with $s = q$, $z_0 = z$, $\nu = 0$ gives the value $q(1 + [Z \neq \emptyset])$. If $\mu \neq 0$ and $k \geq 4$, make $A_F \equiv 1$ and then shift E_F on Z by $d = A_F^2(1 + \chi_F)/\mu$ (Proposition 4.6). This d is ρ-periodic by Lemma 4.5, and since $\mathrm{Tr}(1) = 1$ for odd m it makes $\chi_F \equiv 1$. Then Theorem 5.2 gives $n_{2q} = q^{k-1}/2$ and the value $2q$. □

Proof of Theorem C. Let $k \geq 4$. If A_F has a zero then $\Delta_{\mathrm{exc}} \geq q^2 > 4q$ by Theorem 5.3(iii). If A_F has no zero, Theorem 5.2 gives the full spectrum: for $\mu = 0$ the maximum is the minimax value, and for $\mu \neq 0$ it is $2q$ if $\nu = 0$ and $4q$ otherwise. Let $k = 3$ and $\mu = 0$. If $A_F(a) = 0$, a β with $c_a = 0$ gives $q \cdot q = q^2$ solutions in the class of a, more than $2q$. If A_F has no zero, Theorem 5.1(ii) gives the minimax value. Let $k = 3$ and $\mu \neq 0$. By Theorem 5.1(i) the maximum is $4q$ if $\nu > 0$, otherwise $2q$ if $n_{2q} > 0$, otherwise q. If $Z = \emptyset$, then $n_{2q} = (q - s)q/2$ vanishes if and only if $A_F \equiv 0$. If $Z \neq \emptyset$, the value $2q$ is attained if and only if $\nu = 0$. □

Example 6.1. A minimizer at $k = 3$ need not satisfy the middle-layer condition of Theorem 8.1 in Section 8. For $m = 5$ (field elements written as integers, bit i = coefficient of t^i, $t^5 = t^2 + 1$) let $F_1 = c_0 + 2X^2$, where c_0 is a lift with $A \equiv 1$ of a representative with $\rho = 21$, $\mu = 28$ and $Z = \{12, 25\}$ (listed in Online Resource 1). Then $A_{F_1} \equiv 1$ and $\chi_{F_1} \equiv 1$, and all 1024 exceptional rows of F_1 consist of 512 entries equal to $2q = 64$.

Optimality across levels. Let $k \geq 4$, let $F_k \in R_k[X]$ be a lift and let F_3 be its coefficientwise reduction modulo 8, a *compatible* lift at level 3. Every F_3 has compatible lifts at every level.

Lemma 6.2. *$A_{F_k} = A_{F_3}$, $E_{F_k} = E_{F_3}$ on Z, $\chi_{F_k} = \chi_{F_3}$, and Λ and μ coincide.*

Proof. Reduction $\psi : R_k \to R_3$ is a ring homomorphism compatible with the residue maps. Apply ψ to the representatives in the definitions of A_F and E_F. Then $\psi(2g) = 2\psi(g)$ and $\psi(4c) = 4\psi(c)$, and by Lemmas 4.1 and 4.3 the residues do not depend on the representatives. □

Theorem 6.3. *Let $k \geq 4$ and suppose F_3 minimizes Δ_{exc} at level 3. Then F_k minimizes Δ_{exc} at level k if and only if A_{F_3} has no zero. In particular, if A_{F_3} has no zero, every compatible lift of F_3 is optimal at every level $k \geq 4$.*

Proof. By Lemma 6.2 the invariants agree. Necessity is Theorem C for $k \geq 4$. Conversely, let A have no zero. The level-k criterion asks in addition, when $\mu \neq 0$, that $\chi = 1$ on Z. The level-3 criterion for $\mu \neq 0$ gives $\chi(a) = 1$ at every $a \in Z$ with $A(a) \neq 0$, which is all of Z. When $\mu \neq 0$ and $Z = \emptyset$, a level-3 minimizer has $A \equiv 0$, so both sides of the equivalence fail. □

Corollary 6.4. *A compatible family $(F_k)_{k \geq 3}$ minimizing Δ_{exc} at every level exists if and only if $\mu = 0$ or $Z \neq \emptyset$.*

Proof. If $\mu = 0$, take F_3 with $A \equiv 1$ and apply Theorem 6.3. If $\mu \neq 0$ and $Z \neq \emptyset$, take F_3 with $A \equiv 1$ and $\chi \equiv 1$ on Z, which minimizes at level 3, and apply Theorem 6.3. If $\mu \neq 0$ and $Z = \emptyset$, level 3 requires $A \equiv 0$ and level $k \geq 4$ requires A zero-free, for the same function A by Lemma 6.2. □

Theorem 6.3 and Corollary 6.4 together give Corollary E. The minimax table gives the least value at each level. The two results above say when the same coefficients reach these values at all levels at once, and we keep the two statements separate.

Example 6.5. For $m = 5$ let c_1 be the lift with $A \equiv 0$ of a representative with $\rho = 28$, $\mu = 17$ and $Z = \emptyset$. The same integer coefficients give, over $\mathrm{GR}(8, 5)$, 1024 exceptional rows equal to "32 appears 1024 times" (optimal value q). Over $\mathrm{GR}(16, 5)$ every sampled exceptional row (four rows; the lower bound q^2 holds for all rows by Theorem 5.3(iii)) is "1024 appears 1024 times", whereas the optimum is 64. Lifts with A zero-free behave as predicted by Theorem 6.3.

7 The family $P_{m,r}$ and one-term optimal lifts

Let $m \geq 5$ be odd and $3 \leq r < m$ odd with $\gcd(r, m) = 1$. Put $T_r = \sum_{i<r} X^{2^i}$ and, as in [5, Thm. 3.1],

$$P_{m,r} = \sigma^{m-2} \circ X^3 \circ T_r, \qquad \sigma(x) = x^2,$$

reduced modulo $X^q - X$. With $u = 2^{m-2}$, $v = 2^{m-1}$, $w = 2^{r-2}$ and $T = \sum_{i=0}^{r-3} X^{2^i}$ its reduced form is

$$p = T + X^w + X^w T + X^{w+u} + X^{w+v} + X^u T + X^{u+v},$$

and for $r = 3$ it is $p = X + X^2 + X^3 + X^{1+u} + X^{2+u} + X^{2+v} + X^{u+v}$. The natural lift F reads the same expression with integer coefficients 1. Its degree is $u + v = 3q/4$.

Proposition 7.1 (The residue is APN)**.** *Put $Y(x) = T(x) + x^u + x^v$. On $\mathbb{F}_q$ we have $p(x) = Y(x)^3$, the map Y is additive, and $Y(x) = 0$ only for $x = 0$. Hence p is an APN permutation.*

Proof. On $\mathbb{F}_q$ we have $(x^u)^2 = x^v$ and $(x^v)^2 = x$, and $T^2 = T + x + x^w$. Expanding $(T + x^u + x^v)^3$ and using these relations in characteristic 2 gives the reduced form of p. Additivity is clear. Further $Y^4 = T^4 + x + x^2$, and $T^4 + x + x^2 = T_r(x)$. If $T_r(x) = 0$ then $T_r + T_r^2 = x + x^{2^r}$ gives $x^{2^r} = x$, so $x \in \mathbb{F}_{2^{\gcd(r,m)}} = \mathbb{F}_2$. Since $T_r(1) = r = 1$, we get $x = 0$. For $c \neq 0$ and $d = Y(c) \neq 0$, $p(t + c) - p(t) = dY(t)^2 + d^2 Y(t) + d^3$. Two solutions t, t_0 of $p(t + c) - p(t) = b$ satisfy $d(Y(t) - Y(t_0))(Y(t) + Y(t_0) + d) = 0$, so $t \in \{t_0, t_0 + c\}$. Bijectivity of p follows from that of Y and of $x \mapsto x^3$ for odd m. □

This is the route of [5, Thm. 3.1] (the cube is APN, T_r is invertible, and the outer map is a power of Frobenius), with σ^{m-2} absorbed into Y. The inner power map is the cube; r only indexes the inner linear map T_r (an earlier version of our notes wrongly described $P_{m,r}$ as linearly equivalent to X^{2^r+1}).

Proposition 7.2. *(i) $\rho = 1$: $p' = 1 + X^w + X^u$ for $r \geq 5$ and $p' = 1 + X^2 + X^u$ for $r = 3$, with $\ker K = \{0, 1\}$ and $1 \notin K(\mathbb{F}_q)$. (ii) $\mu = 0$ for $r \geq 5$ and $\mu = 1$ for $r = 3$. (iii) $\Lambda \equiv 0$, so $Z = \mathbb{F}_q$.*

Proof. (i) Only monomials with odd exponent contribute to p'. Further, $x^w = x^u$ implies $x^{2^{m-r}} = x$ after applying Frobenius, so $x \in \mathbb{F}_{2^{\gcd(r,m)}} = \mathbb{F}_2$. Also $K(\mathbb{F}_q) \subseteq \ker \mathrm{Tr}$ while $\mathrm{Tr}(1) = m = 1$. (ii) By Lemma 3.2(i) only the exponents 2, 3 and $2 + 2^j$ contribute to $p^{[2]}$. This gives $p^{[2]} = 1 + X^w + X^u$ for $r \geq 5$ and $1 + X + X^u + X^v$ for $r = 3$. (iii) In characteristic 2, $D_1 X^{2^i} = 1$ and $D_1 X^{2^a + 2^b} = X^{2^a} + X^{2^b} + 1$. Summing these over the monomials of p and adding p' gives 0 (for $r \geq 5$ the number of terms of T is $r - 2$, which is odd). □

Proposition 7.3. *Modulo 4 and on $\mathbb{F}_q$: (i) for $r \geq 5$, $A_F = L + L^2$ with $L = X^{w/2} + X^{u/2}$, and $A_{F+2X^3} = L + L^2 + 1$, which has no zero; (ii) for $r = 3$, $A_F = 1 + X + X^{u/2} + X^u + X^v$, which has no zero, and A is unchanged by $2X^{4+v}$.*

Proof. We use $(X+1)^{2^j} \equiv X^{2^j} + 1 + 2X^{2^{j-1}} \pmod 4$ for $j \geq 1$ (induction on j). For $r \geq 5$, $F' \equiv 1 + 2X + X^w + 2X^{w+1} + X^u + 2X^{u+1} \pmod 4$, and substituting $X+1$ gives the difference $2(X^{w/2} + X^w + X^{u/2} + X^u)$. The term $2X^3$ adds $D_1(X^2) = 1$ by Proposition 4.2(i). Since $\mathrm{Tr}(L+L^2) = 0$ and $\mathrm{Tr}(1) = 1$, the function $L+L^2+1$ has no zero. For $r = 3$ the same computation gives $A_F = 1 + X + X^{u/2} + X^u + X^v$, whose trace is $1 + 4\mathrm{Tr}(X) = 1$. The correction $2J^2$ with $J = X^{2+u}$ keeps A by Proposition 4.6. □

Proposition 7.4. *Let $r = 3$, $Y_0 = X^{u/2}$, $U = X^u$, $V = X^v$. (i) $E_F = X + X^2 + Y_0 + XY_0 + X^2Y_0 + X^2U + V + Y_0V$. (ii) For $F^\sharp = F + 2X^{4+v}$, $E_{F^\sharp} = E_F + X^4 + V + 1$. With $\varphi = 1 + X^2 + XV + Y_0V + UV$ and $A = 1 + X + Y_0 + U + V$ we have $\varphi^2 + A\varphi = E_{F^\sharp} + A^2$ on $\mathbb{F}_q$. Hence $\chi_{F^\sharp} \equiv 1$. (iii) $E_F(0) = 0$ and $A_F(0) = 1$, so $\chi_F(0) = 0$, and the natural lift has $\nu \geq 2$.*

Proof. (i) Modulo 8, $S(x) \equiv 2[F(x) - F(x+1) + F'(x)] + 4F^{[2]}(x)$, so

$$E_F = \overline{(F(X+1) - F(X) - F'(X))/2} + p^{[2]}.$$

The numerator is even because $\Lambda \equiv 0$, and expanding it modulo 4 gives the stated form. (ii) $D_1(X^{2+u}) = X^2 + U + 1$, so $(D_1 j)^2 = X^4 + V + 1$ by Proposition 4.6. The identity is an identity in $\mathbb{F}_2[t]$ after the substitution $Y_0 = t$, $U = t^2$, $V = t^4$, $X = t^8$, which is valid on $\mathbb{F}_q$ because $V^2 = X$. It gives

$$\mathrm{Tr}(\mu E_{F^\sharp}/A^2) = \mathrm{Tr}(1 + (\varphi/A)^2 + \varphi/A) = 1.$$

(iii) Direct evaluation, and χ is ρ-periodic (Lemma 4.5). □

Proof of Theorem A. The correction is divisible by 2, so the residue is unchanged and $F^\sharp$ is a permutation. The correction has degree 3 or $q/2+4$, both below $3q/4$, so $F^\sharp$ keeps the degree $3q/4$ of F. For $r \geq 5$: $\mu = 0$, $Z = \mathbb{F}_q$, and $A_{F^\sharp}$ has no zero (Propositions 7.2 and 7.3). Theorem 5.2 gives $n_{2q} = q^{k-1}/2$ and $n_q = 0$. For $r = 3$: $\mu = 1$, $Z = \mathbb{F}_q$, A has no zero and $\chi \equiv 1$ (Proposition 7.4), so $\nu = 0$ and $n_{2q} = q^{k-1}/2$. Ordinary directions are covered by Lemma 3.3 and Proposition 7.1. Optimality is the row $Z \neq \emptyset$ of Theorem B. For the natural lift with $r = 3$, A_F has no zero, so Theorem 5.2 bounds every entry by $4q$, and Proposition 7.4(iii) gives $n_{4q} > 0$, so $\Delta_{\mathrm{exc}} = 4q$. For $r \geq 5$, $A_F = L + L^2$ vanishes where $L \in \mathbb{F}_2$. Theorem 5.3(iii) and the case $k = 3$, $\mu = 0$ in the proof of Theorem C give $\Delta_{\mathrm{exc}} \geq q^2$.

Cost: we count operations in the *termwise evaluation model*, in which the squares x^{2^i} $(1 \leq i \leq m-1)$ are computed once, each monomial is formed from them by ring multiplications, multiplied by its coefficient when the coefficient is not 1, and the terms are added. The squares are already computed for F. For $r \geq 5$, $x^3 = x \cdot x^2$; for $r = 3$, $x^{4+v} = x^4 \cdot x^{2^{m-1}}$. Each costs one ring multiplication. Doubling and adding cost two additions. Other evaluation circuits of the same polynomial (factored forms, shared products, hardware) have their own costs, which this count does not address. □

Corollary 7.5 (The natural lift at $k = 3$)**.** *Let $r \geq 5$, let F be the natural lift of $P_{m,r}$ over $R_3 = \mathrm{GR}(8, m)$ and fix an exceptional α. Then $n_{2q} = q(q-2)/2$, and either $n_{q^2} = 2$ (if $r \equiv 1 \pmod 4$) or $n_{2q^2} = 1$ (if $r \equiv 3 \pmod 4$). All other positive values have multiplicity zero. In particular $\Delta_{\mathrm{exc}}(F) = q^2$ if $r \equiv 1 \pmod 4$ and $\Delta_{\mathrm{exc}}(F) = 2q^2$ if $r \equiv 3 \pmod 4$.*

Proof. By Propositions 7.2 and 7.3, $\rho = 1$, $\mu = 0$, $Z = \mathbb{F}_q$ and $A_F = L + L^2$ with $L = X^{w/2} + X^{u/2}$. If $L(a) = 0$, raising $a^{2^{r-3}} = a^{2^{m-3}}$ to the power 8 gives $a^{2^r} = a^{2^m} = a$, so $a \in \mathbb{F}_{2^{\gcd(r,m)}} = \mathbb{F}_2$. The value $L(a) = 1$ is impossible, since $\mathrm{Tr}L(a) = \mathrm{Tr}(a) + \mathrm{Tr}(a) = 0$ while $\mathrm{Tr}(1) = 1$. Hence $A_F(a) = 0$ exactly for $a \in \{0, 1\}$.

Fix a pair $\{a, a+1\}$ and β with $\overline{\beta} = D_1 p(a)$; only this pair contributes (Lemma 3.2(iii)). Since $\Lambda \equiv 0$, both classes pass the gate for the same $\beta \bmod 4$ (Lemma 3.4), and $c_{a+1} = c_a + E_F(a)$ (Lemma 3.5(iii)). If $a \notin \{0,1\}$, then $L_a(y) = A_F(a)y$ is bijective, and by Lemma 3.5(ii) each class contributes q for each of the q values of c_a. This gives q values of β with entry $2q$ for each of the $(q-2)/2$ such pairs. For the pair $\{0,1\}$, $L_0 \equiv 0$, so the class of a contributes $q \cdot q = q^2$ if $c_a = 0$ and nothing otherwise. With the lifts $x = 0$, $\tau = 1$ in Lemma 4.3, $E_F(0)$ is the residue of $(F(2) - 2F(1) + F(0))/4$. The natural lift has $3r - 2$ monomials, one with exponent 1, one with exponent 2 and all others with exponent at least 4, so $F(1) = 3r - 2$, $F(2) \equiv 6 \pmod 8$ and $F(0) = 0$. Thus $F(2) - 2F(1) + F(0) \equiv 10 - 6r \pmod 8$, and $E_F(0) = (5 - 3r)/2 \bmod 2$, which is 1 for $r \equiv 1 \pmod 4$ and 0 for $r \equiv 3 \pmod 4$. In the first case $c_0 = 0$ and $c_0 = 1$ each give one entry q^2; in the second case $c_0 = 0$ gives one entry $2q^2$. As a check, $2q \cdot q(q-2)/2 + 2q^2 = q^3$. □

Corollary 7.5, multiplicities included, is formalized (Appendix A). It agrees with the full computation of all 16384 exceptional rows of the natural lift of $P_{7,5}$ ($n_{2q} = 8064$, $n_{q^2} = 2$). For $r \equiv 3 \pmod 4$ we checked sampled rows of the natural lift of $P_{9,7}$ over $\mathrm{GR}(8,9)$ (Appendix A).

For $r \geq 5$ the lift $F^\sharp$ also optimizes the $k = 3$ middle layer and makes the penultimate layer optimally flat (Theorem F, $\mu = 0$). For $r = 3$ we have $\mu = 1$, so an optimal middle layer would need $A \equiv 0$, whereas $A_{F^\sharp}$ has no zero (Theorem C).

8 The lower layers

Theorem 8.1 (Middle layer, $k = 3$)**.** *Let $k = 3$ and $\alpha = 2u$ with u a unit. Then $F(x+\alpha) - F(x)$ depends only on $\overline{x}$, and residues a and b give the same value if and only if $b = a$, or $b = a + \rho$ and $\overline{u}A_F(a) + \overline{u}^2\mu = 0$. Hence all nonzero entries of every row in v_1 equal q^2 if and only if $A_F \equiv 0$ when $\mu \neq 0$, and A_F has no zero when $\mu = 0$.*

Proof. $F(x+2u) - F(x) = 2uF'(x) + 4u^2F^{[2]}(x) + 8(\cdots)$. Both terms depend only on $\overline{x}$ modulo 8 (Lemma 4.1(i)). Modulo 4 the value is the lift of $2\overline{u}p'(a)$, so equal values force $b \in \{a, a+\rho\}$. For $b = a + \rho$ and $x_b = x_a + \tau$ the difference is $\equiv 4(\overline{u}A_F(a) + \overline{u}^2\mu) \pmod 8$. As $\overline{u}$ runs over $\mathbb{F}_q^*$, the condition "never zero" means $A = 0$ if $\mu \neq 0$ and $A \neq 0$ if $\mu = 0$. □

Theorem 8.2 (Penultimate layer)**.** *Let $k \geq 4$ and $\alpha \in 2^{k-2}R_k \setminus 2^{k-1}R_k$. Every row has $q - s$ entries q^{k-1} and $s/2$ entries $2q^{k-1}$. In particular the penultimate layer is optimally flat (all nonzero entries equal to q^{k-1}) if and only if A_F has no zero.*

Proof. $\alpha^2 = 0$, so $F(x+\alpha) - F(x) = \alpha F'(x)$, which depends only on $F'(x) \bmod 4$ and hence only on $\overline{x}$. Two residues give the same value if and only if $F'(x_a) \equiv F'(x_b) \pmod 4$. This forces $b \in \{a, a+\rho\}$, and for $b = a + \rho$ it holds if and only if $A_F(a) = 0$. □

Corollary 8.3. *(i) If $\mu \neq 0$, middle-layer optimality at $k = 3$ ($A \equiv 0$) and exceptional optimality at $k \geq 4$ (A zero-free) are incompatible. A lift with $A \equiv 0$ has $\Delta_{\mathrm{exc}} \geq q^2$ and a penultimate layer whose nonzero entries all equal $2q^{k-1}$, at every $k \geq 4$. (ii) If $\mu = 0$, a zero-free A gives the middle layer at $k = 3$, exceptional optimality at all k and an optimally flat penultimate layer at all $k \geq 4$. (iii) If $\mu \neq 0$ and $Z = \emptyset$, the optimal exceptional value itself rises from q at $k = 3$ to $2q$ at $k \geq 4$. This change of value is a separate phenomenon from the incompatibility in (i).*

This proves Theorem F.

Intermediate layers. For $1 \le j \le k-3$ the layers v_j lie between the middle layer of $k=3$ and the penultimate layer. Two further invariants of p enter: $c_3 = [X^3]p$ and the affine map $\pi = p^{[2]}$ of Lemma 3.2(i). For $\alpha = 2^j u$ with u a unit put

$$w_\alpha = \pi + [j=1]\,\overline{u}\,c_3, \qquad B_\alpha = A_F + [j=1]\,\overline{u}\,\mu,$$
$$s_\alpha = \#B_\alpha^{-1}(0), \qquad d = k-j-2.$$

Theorem 8.4 (Intermediate layers)**.** *Let $k \ge 4$, $1 \le j \le k-3$ and $\alpha = 2^j u$ with u a unit.*

(i) Every nonzero entry of the row of α is a multiple of q^{j+1}.

(ii) If $c_3 = 0$ and π has no zero, the row has $(q-s_\alpha)q^d$ entries q^{j+1}, $\frac{s_\alpha}{2}q^d$ entries $2q^{j+1}$, and no other nonzero entries.

(iii) All nonzero entries of all rows of v_j equal q^{j+1} if and only if $c_3 = 0$, π has no zero, and A_F has no zero (when $j \ge 2$ or $\mu = 0$), respectively $A_F \equiv 0$ (when $j = 1$ and $\mu \ne 0$).

Proof. Put $n = k-j$. Then $F(x+\alpha) - F(x) = \alpha T(x)$ with $T = \sum_{i\ge1} \alpha^{i-1} F^{[i]}$, and $\alpha y = \alpha y'$ if and only if $y \equiv y' \pmod{2^n}$. Hence $\delta_F(\alpha,\beta) = 0$ unless $\beta = \alpha\gamma$, and then $\delta_F(\alpha,\alpha\gamma) = q^j\,\#\{y \in R_k/2^nR_k : T(y) \equiv \gamma\}$. Since $\overline{T} = p'$ is two-to-one on the pairs $\{a, a+\rho\}$, solutions lie in one pair. The derivative $T' = 2F^{[2]} + 3\alpha F^{[3]} + \cdots$ has coefficients in $2R_k$; with $W = T'/2$ we get $T(x+2z) = T(x) + 4\Phi_x(z)$, where $\Phi_x(z) = W(x)z + T^{[2]}(x)z^2 + \sum_{r\ge3} 2^{r-2}T^{[r]}(x)z^r$ has residue $w_\alpha(a)y + c_3y^2$ for $\overline{x} = a$ (the residue of W is $\pi + [j=1]\overline{u}c_3$ and that of $T^{[2]}$ is $p^{[3]} = c_3$; the only exponent $2^a + 2^b$ with $\binom{e}{3}$ odd is 3). On the class of a, $T(x_a + 2z) = \gamma$ means $\gamma \equiv T(x_a) \pmod 4$ and $\Phi_{x_a}(z) \equiv g \pmod{2^d}$ with $g = (\gamma - T(x_a))/4$; the second condition depends on z modulo 2^d, and each such z has q lifts modulo 2^{n-1}. So the class contributes $q^{j+1}N_a(g)$ with $N_a(g) = \#\{z \bmod 2^d : \Phi_{x_a}(z) \equiv g\}$, which proves (i). Moreover $T(x+\tau) - T(x) \equiv 2[A_F(a)] + [j=1]\,2[\overline{u}\mu] = 2[B_\alpha(a)] \pmod 4$, so the values of γ reached from the two classes of a pair lie in the same residue class modulo 4 if $B_\alpha(a) = 0$ and in different residue classes modulo 4 otherwise; in the second case the two classes reach disjoint sets of values.

(ii) Here $w_\alpha = \pi$ has no zero and $c_3 = 0$, so Φ_{x_a} has unit derivative and every g has exactly one solution by Hensel's lemma: $N_a \equiv 1$ on all q^d values of g. So each class of a pair reaches every γ of its residue class modulo 4, each with the entry q^{j+1}. A pair with $B_\alpha \ne 0$ therefore gives $2q^d$ entries q^{j+1}. For a pair with $B_\alpha = 0$ both classes reach every γ of the same residue class modulo 4, so the pair gives q^d entries $2q^{j+1}$.

(iii) If $c_3 \ne 0$, then w_α is affine with nonzero linear part, so $w_\alpha(a) \ne 0$ for some a, and $\#\{y : w_\alpha(a)y + c_3y^2 = \overline{g}\} \in \{0,2\}$ gives entries $2q^{j+1}$ or more. If $c_3 = 0$ and $\pi(a) = 0$, then $\Phi_{x_a} \equiv 0 \pmod 2$, so its q^d values fall into at most q^{d-1} classes and some $N_a(g) \ge q$. Otherwise (ii) applies, and the row is flat if and only if $s_\alpha = 0$. For $j \ge 2$ or $\mu = 0$, $B_\alpha = A_F$. For $j = 1$ and $\mu \ne 0$, B_α has no zero for every unit $\overline{u}$ if and only if $A_F(a)/\mu \notin \mathbb{F}_q^*$ for all a, that is $A_F \equiv 0$. □

Corollary 8.5. *Let $r \ge 5$ in Theorem A. For every $k \ge 3$ and every $1 \le j \le k-2$, every row of v_j of $F^\sharp$ has all nonzero entries equal to q^{j+1}, the least possible value. The natural lift has, in every row of v_j, $(q-2)q^{k-j-2}$ entries q^{j+1} and q^{k-j-2} entries $2q^{j+1}$.*

Proof. For $r \ge 5$, p has no monomial X^3, so $c_3 = 0$, and $\pi = 1 + X^w + X^u$ (Proposition 7.2(ii)) has no zero because $\mathrm{Tr}\pi(a) = \mathrm{Tr}(1) = 1$. Also $\mu = 0$, $A_{F^\sharp}$ has no zero and $A_F^{-1}(0) = \{0,1\}$ (Corollary 7.5). Apply Theorem 8.4 for $j \le k-3$, Theorem 8.2 for $j = k-2$ and Theorem 8.1 for $k = 3$, $j = 1$. Minimality is Theorem 8.4(i) for $j \le k-3$, Theorem 8.2 for $j = k-2$, and Theorem 8.1 for $k = 3$, where the difference is constant on residue classes of q^2 points. □

Thus for $r \geq 5$ the one-term lift is optimal on every layer except the top layer, whose value $2q^{k-1}$ is forced for every lift [1]. When $c_3 \neq 0$ or π has a zero, the class counts N_a are given by a Hensel tree whose shape depends on whether $w_\alpha(a) = 0$ and whether $c_3 = 0$. A residue with $w_\alpha(a) = 0 \neq c_3$ produces a universal multiset of values, from which every row of v_j meeting such a residue has an entry at least $q^{j+1+\lfloor (k-j-2)/2 \rfloor}$. For $r = 3$ ($c_3 = 1$, $\pi(1) = 0$, and $X + X^u + X^v$ is bijective because $3 \nmid m$) this happens for every lift.

The finer multisets in these cases are governed by one elementary recursion. For $\Phi \in R_k[z]$ and $D \geq 0$ let $N_\Phi^D(g) = \#\{z \in R_k/2^D R_k : \Phi(z) \equiv g \pmod{2^D}\}$; the class contribution in the proof of Theorem 8.4 is $q^{j+1} N^d_{\Phi_{x_a}}$.

Proposition 8.6 (Digit recursion)**.** *Let $\Phi \in R_k[z]$, $2 \leq D \leq k$ and $\hat{y} \in R_k$ a lift of $y \in \mathbb{F}_q$. If $\Phi'(\hat{y})$ is a unit, then for every g with $\overline{g} = \overline{\Phi}(y)$ exactly one $z \equiv \hat{y} \pmod 2$, $z \bmod 2^D$, satisfies $\Phi(z) \equiv g$. If $\Phi'(\hat{y}) \in 2R_k$, then $\Phi_y(z) = (\Phi(\hat{y} + 2z) - \Phi(\hat{y}))/4$ lies in $R_k[z]$, and*

$$\#\{z \equiv \hat{y} \ (2) : \Phi(z) \equiv g \ (2^D)\} = q\,[\, g \equiv \Phi(\hat{y}) \ (4)\,]\ N^{D-2}_{\Phi_y}((g - \Phi(\hat{y}))/4),$$

for any choice of the quotients by 4; the count does not depend on the choice since $D - 2 \leq k - 2$.

Proof. If $\Phi'(\hat{y})$ is a unit, $\Phi(z) - \Phi(z') = (z - z')U$ with $U \equiv \Phi'(\hat{y}) \pmod 2$ for $z \equiv z' \equiv \hat{y}$, so Φ is injective from $\hat{y} + 2R_k/2^D$ into $\Phi(\hat{y}) + 2R_k/2^D$, two sets of q^{D-1} elements. Otherwise $\Phi(\hat{y}+2z) - \Phi(\hat{y}) = \sum_{r \geq 1} 2^r \Phi^{[r]}(\hat{y}) z^r$ has all coefficients in $4R_k$, and $\Phi(\hat{y}+2z) \equiv g \pmod{2^D}$ holds if and only if $g \equiv \Phi(\hat{y}) \pmod 4$ and $\Phi_y(z) \equiv (g - \Phi(\hat{y}))/4 \pmod{2^{D-2}}$, a condition on z modulo 2^{D-2}; each such z has q lifts modulo 2^{D-1}. □

Iterating Proposition 8.6 over the residue classes determines every row of every intermediate layer of every lift, after at most $\lceil d/2 \rceil$ steps and with $O(q)$ evaluations of T per critical point. The notes on the intermediate layers in Online Resource 1 classify the resulting multisets. Every configuration of a single class or a merged pair is given in closed form in terms of level data of T, except a few configurations of merged or deep classes, listed in those notes, that need at least five independent conditions on the Taylor data of F; there the recursion still gives the answer. In particular, for $j \leq k - 4$, a class with $c_3 = 0 = \pi(a)$ for which the coefficients of z and z^2 in Φ_{x_a} lie in $4R_k$ has an entry at least q^{j+3}. All closed forms and the recursion agree with exhaustive computation for $m = 5$ and $d \leq 5$ (133 configurations, 75 full rows, and 691 further rows from earlier computations).

9 Cryptographic consequences

Write $P_F(\alpha \to \beta) = \delta_F(\alpha, \beta)/q^k$. For $F^\sharp$ the maximum over unit α is $2q/q^k = 2/q^{k-1}$, and over ordinary α at most $2/q^k$. For a general p and an optimal lift the exceptional maximum is the minimax value divided by q^k.

Proposition 9.1 (Valuation preservation)**.** *If p satisfies (H) and F is a lift of p, then $v_2(F(x + \alpha) - F(x)) = v_2(\alpha)$ for every $\alpha \neq 0$ and every x.*

Proof. For a unit α the residue difference is nonzero since p is a permutation. For $\alpha = 2^s u$ with $s \geq 1$, $F(x + \alpha) - F(x) = \alpha(F'(x) + \alpha H(x, \alpha))$, and the bracket is a unit since p' has no zero. □

The probability space. Let the state be $x_0 = X_0 \in R_k^n$, uniform and independent of independent uniform round keys $\kappa_1, \ldots, \kappa_N \in R_k^n$, and $x_i = M \cdot S(x_{i-1}) + \kappa_i$. Equivalently, take a fixed plaintext and add an independent uniform whitening key κ_0. For a characteristic $\Omega = (\Delta_0 \to \cdots \to \Delta_N)$ let $\Pr[\Omega]$ be the probability, over $(X_0, \kappa_1, \ldots, \kappa_N)$, that the pair $(x_0, x_0 + \Delta_0)$ follows Ω.

Example 9.2 (A fixed plaintext breaks the product formula)**.** Take $\mathrm{GR}(8,5)$, $S = F^\sharp$ for $P_{5,3}$, $n = 1$, $M = 1$ and the fixed pair $(0, 1)$. Then $S(1) - S(0) = 1$, so the first transition holds for every key. The second input $x_1 = S(0) + \kappa_1$ is uniform, and the key-averaged probability of $1 \to 1 \to 1$ is $\delta(1,1)/|R| = 64/32768 = 1/512$. The product of transition probabilities is $(1/512)^2$.

Proposition 9.3 and part (a) of Corollary G are the Markov-cipher argument of Lai, Massey and Murphy [9] and the branch-number argument of Daemen and Rijmen [10, 11], written for keys added in R_k and combined with the valuation preservation above, and Proposition 9.5 below is the two-round bound of Hong et al. [12] (see also [13]) transferred to ring addition. What is new is the per-S-box input: the sharp unit-direction value of Theorems A and B, the layer values of Section 8, the transfer of the branch number of $\overline{M}$ to R_k^n, and, in Proposition 9.6, the observation that valuation preservation fixes the active words of a layer at both ends of a two-round differential, so that the layer values of a lift enter the differential bound without a combinatorial factor. Throughout, only the additivity of M and the invertibility of $\overline{M}$ are used.

Proposition 9.3 (Markov property)**.** *In the probability space above,* $\Pr[\Omega] = \prod_{i=1}^{N} P_S(\Delta_{i-1} \to M^{-1}\Delta_i)$*, where*

$$P_S(a \to b) = \prod_{j=1}^{n} \frac{\delta_F(a_j, b_j)}{q^k} = q^{-kn} \prod_{j=1}^{n} \delta_F(a_j, b_j)$$

is the probability that a uniform $x \in R_k^n$ *satisfies* $S(x + a) - S(x) = b$*.*

Proof. Let E_i be the event $S(x_{i-1} + \Delta_{i-1}) - S(x_{i-1}) = M^{-1}\Delta_i$. The events $E_1, \ldots, E_{i-1}$ are determined by $(X_0, \kappa_1, \ldots, \kappa_{i-2})$. The input $x_0 = X_0$ is uniform, and for $i \geq 2$, $x_{i-1} = MS(x_{i-2}) + \kappa_{i-1}$ with κ_{i-1} uniform and independent of $(X_0, \kappa_1, \ldots, \kappa_{i-2})$. So x_{i-1} is uniform conditionally on $E_1 \cap \cdots \cap E_{i-1}$. Hence $\Pr[E_i \mid E_1 \cap \cdots \cap E_{i-1}] = P_S(\Delta_{i-1} \to M^{-1}\Delta_i)$. □

Proof of Corollary G. Let $\Pr[\Omega] > 0$ and let $\overline{\Delta}_i$ and u_i be the reductions modulo 2 of Δ_i and of $M^{-1}\Delta_i$. By Proposition 9.1, $\mathrm{supp}(u_i) = \mathrm{supp}(\overline{\Delta}_{i-1})$, and $\overline{\Delta}_i = \overline{M}u_i$. With $a_i = \mathrm{wt}(\overline{\Delta}_{i-1})$, the number of unit-active S-boxes in round i, we get $a_i + a_{i+1} = \mathrm{wt}(u_i) + \mathrm{wt}(\overline{M}u_i) \geq B$. Since $\overline{\Delta}_0 \neq 0$, all $u_i \neq 0$, and $\sum_i a_i \geq B\lfloor N/2 \rfloor$. Each unit-active S-box contributes at most $2/q^{k-1}$, and the others at most 1. This proves (a). Part (b) is Proposition 9.4, part (c) is Propositions 9.5 and 9.6, with $D_0 = 2q$ from Theorem A and D_s from Proposition 9.4, and part (d) is Proposition 9.7. □

For example, with $m = 7$, $k = 3$ (a 21-bit S-box), $n = 4$, an MDS matrix $\overline{M}$ ($B = 5$) and $N = 8$, this bound is 2^{-260} for $F^\sharp$ of $P_{7,5}$. For the natural lift of $P_{7,5}$ the worst exceptional entry at $k = 3$ is exactly q^2 (Corollary 7.5; all 16384 exceptional rows were also computed, Appendix A), so the same argument gives only 2^{-140}, and no better bound per unit-active S-box is possible with this method.

Lower layers. Every nonzero difference $\Delta \in R_k^n$ has a layer: the largest s with $\Delta \in (2^s R_k)^n$. Then $s \leq k-1$ and some word of Δ lies outside $2^{s+1}R_k$. For $s < k$ the digit $\langle 2^s y \rangle_s = \overline{y}$ of an element of $2^s R_k$ is well defined, since $2^s(y - y') = 0$ forces $y - y' \in 2^{k-s}R_k \subseteq 2R_k$. The branch-number argument works in every layer.

Proposition 9.4 (Characteristics in every layer)**.** *In the setting of Corollary G, let F be any lift of p and $1 \le s \le k-1$, let the input difference Δ_0 have layer s, and let D_s be the largest entry of the rows $\delta_F(2^s u, \cdot)$, u a unit. Then $\Pr[\Omega] \le (D_s/q^k)^{B\lfloor N/2 \rfloor}$. For every lift $D_{k-1} = 2q^{k-1}$, and for $F^\sharp$ with $r \ge 5$, $D_s = q^{s+1}$ for $1 \le s \le k-2$.*

Proof. Let $\Pr[\Omega] > 0$. A polynomial maps a difference in $2^s R_k$ to a difference in $2^s R_k$, since $x - y$ divides $F(x) - F(y)$, and M maps $(2^s R_k)^n$ into itself because it is additive. So every Δ_i and every S-box output difference lies in $(2^s R_k)^n$. By Proposition 9.1 a word of an S-box input difference has nonzero digit if and only if the corresponding output word has, and $\langle M(2^s y) \rangle_s = \overline{M}\,\overline{y}$ because $M(2^s y) = 2^s M(y)$. Since $\overline{M}$ is invertible, the digit vectors stay nonzero, so the layer s is invariant along Ω. With a_i the number of words of Δ_{i-1} with nonzero digit we get $a_i + a_{i+1} \ge B$ exactly as in the proof of Corollary G(a), and each such word contributes at most D_s/q^k. For every lift, $F(x + 2^{k-1}u) - F(x) = 2^{k-1} u F'(x)$ depends on $\overline{x}$ only through $\overline{u}\,p'(\overline{x})$, and p' is two-to-one, so every entry of a top-layer row is 0 or $2q^{k-1}$. For $F^\sharp$ and $s \le k-2$ the value is Corollary 8.5. □

In the example above ($m = 7$, $k = 3$, $n = 4$, $B = 5$, $N = 8$) the layer $s = 1$ gives 2^{-140} for $F^\sharp$ but only 2^{-120} for the natural lift, whose middle-layer rows contain the value $2q^2$ because A_F vanishes at 0 and 1 (Theorem 8.1). The top layer $s = 2$ gives 2^{-120} for every lift. So, for $r \ge 5$, the one-term lift improves the bound in every layer below the top, and the weakest of these bounds is the top-layer one, which is the same for every lift.

Differentials and fixed keys. A differential is the union of all characteristics with given input and output differences. For $x \in R_k^n$ let $\mathrm{wt}(x)$ be the number of nonzero words.

Proposition 9.5 (Two-round differentials)**.** *Let M be an additive automorphism of R_k^n whose reduction $\overline{M}$ has branch number B, let S apply a permutation G of R_k to each word, and let $\eta = \max\{\delta_G(\alpha, \beta)/q^k : \alpha \ne 0\}$. For every $a \ne 0$ and every b, the probability, over a uniform x_0 and independent uniform κ_1, κ_2, that the pair $(x_0, x_0 + a)$ has output difference b after the rounds $x \mapsto M \cdot S(x) + \kappa_i$, $i = 1, 2$, is at most η^{B-1}. The same bound holds after any number $N \ge 2$ of rounds with independent uniform keys.*

Proof. First, $\mathrm{wt}(x) + \mathrm{wt}(Mx) \ge B$ for every nonzero $x \in R_k^n$: if s is the layer of x and $x = 2^s y$, then $\overline{y} \ne 0$, $\mathrm{wt}(x) \ge \mathrm{wt}(\overline{y})$ and $\mathrm{wt}(Mx) \ge \mathrm{wt}(\overline{M}\overline{y})$, because $2^s v \ne 0$ for a unit v. Second, summing over the difference x after the first S-box layer, the key κ_1 makes the second input uniform, so the probability is $\sum_x P_S(a \to x)\, P_S(Mx \to c)$ with $c = M^{-1}b$ and P_S as in Proposition 9.3. Since G is a bijection, a term is nonzero only if x and Mx have the supports I of a and J of c. If some term is nonzero, then $|I| + |J| \ge B$; choose $I' \subseteq I$ and $J' \subseteq J$ with $|I'| + |J'| = B - 1$ and bound the factors indexed by I' and J' by η. Two contributing x with the same coordinates on $I \setminus I'$, whose images under M have the same coordinates on $J \setminus J'$, differ by some e with $\mathrm{wt}(e) + \mathrm{wt}(Me) \le B - 1$, so $e = 0$. The remaining sum is therefore at most $\prod_{j \in I \setminus I'} \sum_v \delta_G(a_j, v)/q^k \cdot \prod_{j \in J \setminus J'} \sum_u \delta_G(u, c_j)/q^k = 1$, since the rows and the columns of the difference table of a permutation sum to q^k. For $N \ge 2$ rounds condition on the difference c' after $N - 2$ rounds. It is a function of x_0 and the first $N - 3$ keys, so by the argument of Proposition 9.3 the state after $N - 2$ rounds is uniform and independent of c' and of the last two keys. Moreover $c' \ne 0$ because every round is a bijection. So the probability is an average of two-round probabilities. □

For every lift F of p the value η is $2/q$: the solutions counted by $\delta_F(\alpha, \beta)$, $\alpha \ne 0$, lie in at most two residue classes, and the top layer attains $2q^{k-1}$. With an MDS layer ($B = n + 1$) every differential therefore has probability at most $(2/q)^n$, for instance 2^{-24} for $m = 7$, $n = 4$. This bound

does not see the coefficient choice. Below the top layer it can be separated. For $0 \le s \le k-1$ let D_s be the largest entry of the rows $\delta_F(2^s u, \cdot)$, u a unit; so D_0 is the largest entry on unit input differences and D_s for $s \ge 1$ is the value of Proposition 9.4.

Proposition 9.6 (Differentials in a layer)**.** *In Proposition 9.5 let $G = F$ be a lift of p and let $a \neq 0$ have layer s, $0 \le s \le k-1$. Then for every b the probability is at most $(D_s/q^k)^{B-1}$, after two or more rounds.*

Proof. Put $c = M^{-1}b$ and $P(u,v) = \delta_F(u,v)/q^k$. As in the proof of Proposition 9.5 the probability is $\sum_x \prod_i P(a_i, x_i) \prod_j P((Mx)_j, c_j)$. In a nonzero term $v_2(x_i) = v_2(a_i)$ and $v_2((Mx)_j) = v_2(c_j)$ for all i and j, with $v_2(0) = k$ (Proposition 9.1). All words of a, hence of x and of Mx, lie in $2^s R_k$. So in every nonzero term the supports of the digit vectors $\langle x \rangle_s$ and $\langle Mx \rangle_s = \overline{M}\langle x \rangle_s$ are the sets

$$U = \{i : \ v_2(a_i) = s\}, \qquad V = \{j : \ v_2(c_j) = s\},$$

which depend only on a and b, not on x. Since $U \neq \emptyset$ and $\overline{M}$ has branch number B, a nonzero term forces $|U| + |V| \ge B$. Choose $I' \subseteq U$ and $J' \subseteq V$ with $|I'| + |J'| = B - 1$. In every nonzero term the factors indexed by I' and J' have inputs of exact valuation s and are at most D_s/q^k, and the rest of the proof of Proposition 9.5 applies with η replaced by D_s/q^k. For $N \ge 2$ rounds, the difference after $N-2$ rounds again has layer s, since valuations are kept word by word and $\overline{M}$ is invertible, and the conditioning argument of Proposition 9.5 applies. □

The proof uses only that F is a bijection with $v_2(F(x+\alpha) - F(x)) = v_2(\alpha)$ for all x and $\alpha \neq 0$.

The fixed output difference is what makes the sets U and V independent of x. Without valuation preservation one would have to split the sum according to the support of $\langle Mx \rangle_s$, at the price of a factor up to 2^n. In the example ($m = 7$, $k = 3$, $n = 4$, $B = 5$) the bounds of Proposition 9.6 are:

	$s = 0$ (unit word)	$s = 1$	$s = 2$ (top)
$F^\sharp$ of $P_{7,5}$	$(2/q^2)^4 = 2^{-52}$	$(1/q)^4 = 2^{-28}$	$(2/q)^4 = 2^{-24}$
natural lift of $P_{7,5}$	$(1/q)^4 = 2^{-28}$	$(2/q)^4 = 2^{-24}$	$(2/q)^4 = 2^{-24}$

They use $D_0 = 2q$ for $F^\sharp$ (Theorem A) and $D_0 = q^2$ for the natural lift (Corollary 7.5 and Lemma 3.3), $D_1 = q^2$ and $2q^2$ (Corollary 8.5), and $D_2 = 2q^2$ for every lift (Proposition 9.4). So the coefficient choice changes the differential bound in every layer below the top. These are bounds for single differentials (a, b); the characteristic bounds 2^{-260} and 2^{-140} above are a different quantity, and the bound over all differentials stays 2^{-24}.

Proposition 9.7 (Fixed keys)**.** *In the setting of Proposition 9.5, fix $a \neq 0$ and b. For a key pair $\kappa = (\kappa_1, \kappa_2)$ let P_κ be the proportion of x_0 for which $(x_0, x_0 + a)$ has output difference b; it depends only on κ_1. The average of P_κ over all key pairs is the probability of Proposition 9.5. Hence, for every $t > 0$, at most a proportion η^{B-1}/t of the key pairs satisfies $P_\kappa \ge t$, and at most $(D_s/q^k)^{B-1}/t$ under the hypotheses of Proposition 9.6.*

Proof. Averaging over κ counts the triples $(x_0, \kappa_1, \kappa_2)$, which gives the first statement. The second is Markov's inequality. □

Let a fixed differential have average probability at most ε, for instance $\varepsilon = \eta^{B-1}$ or $\varepsilon = (D_s/q^k)^{B-1}$, and let $0 < \delta < 1$. Taking $t = \varepsilon/\delta$, all key pairs outside a set of proportion at most δ give this differential probability less than ε/δ. So a small average bound gives, for each fixed differential, a bound valid for all key pairs outside a small proportion. It does not give the bound ε itself for each key, and it is not a simultaneous assertion for all differentials.

The residue partition. The following elementary fact fixes what the bound can and cannot certify.

Proposition 9.8 (Residue partition)**.** *Let E be the map $x \mapsto x + \kappa_0$ followed by N rounds $x \mapsto M \cdot S(x) + \kappa_i$ on R_k^n, where S applies any polynomial $G \in R_k[X]$ to each word and M is any additive map of R_k^n. For every choice of keys, $x \equiv y \pmod{2R_k^n}$ implies $E(x) \equiv E(y) \pmod{2R_k^n}$. Hence, for two distinct inputs in the same residue class, the outputs lie in the same residue class with probability 1 for every key, while for a uniformly random permutation of R_k^n this probability is $(q^{(k-1)n} - 1)/(q^{kn} - 1) < q^{-n}$.*

Proof. In R_k, $x - y$ divides $G(x) - G(y)$, so $2 \mid x - y$ implies $2 \mid G(x) - G(y)$ in every word. An additive map sends $2R_k^n$ into itself, and a common key cancels in the difference. For a random permutation, given the image of x the image of y is uniform among the other $q^{kn} - 1$ elements, and $q^{(k-1)n} - 1$ of them lie in the residue class of the image of x. □

The event of Proposition 9.8 is the union of the $q^{(k-1)n} - 1$ nonzero output differences in $2R_k^n$, each of which obeys the per-differential bounds of Corollary G, so there is no contradiction. It is caused by the polynomial structure itself, for every S-box and every coefficient choice, in line with the observation of [1] that non-polynomial maps may be preferable. This is a structural property, not a key-recovery attack. It says that a complete cipher must contain a component that does not preserve congruence modulo 2: a non-polynomial permutation of R_k or a linear layer that feeds higher 2-adic digits into the residue, such as the digit exchange over $\mathrm{GR}(4, m)$ studied in the author's separate manuscript mentioned in the introduction. Not every change of group suffices: a key added by XOR on the binary digits of the coefficients preserves the partition, since the lowest digit of the sum depends only on the lowest digits. Our results then apply to the segments built from ring operations. Inside such a segment each unit-active S-box is crossed with probability at most $2/q^{k-1}$ for $F^\sharp$, against $4/q^{k-1}$ or q^{2-k} and more for the natural lift. If the segment starts from a uniform state, adds an independent uniform key in R_k in each of its rounds and is entered by a difference with a unit component, Proposition 9.3 and Corollary G, with N the number of rounds of the segment, bound characteristics through it; a bound for the whole cipher would in addition need a Markov property of the other components. The positive content is therefore a design criterion with a sharp target (Theorem B), a certificate that is checkable modulo 8 (Theorem D and Corollary E), and, for $P_{m,r}$, an optimal lift at the price of one multiplication (Theorem A). It is not a security claim for a complete cipher.

10 Representatives, applications and open problems

Representatives with $\mu = 0$ and $Z = \emptyset$. The best cell of the minimax table needs a representative with $\mu = 0$ and $Z = \emptyset$. Random searches over $p = L(A(x)^3)$ with $\mathbb{F}_2$-linear A, L find such representatives for $m = 5, 7, 9, 11$ (for $m = 7$, 238 among $2 \cdot 10^6$ trials; for $m = 9$, 81 among $3 \cdot 10^6$; for $m = 11$, 7 among 10^6). The following family settles their existence for every odd $m \geq 5$.

Theorem 10.1. *Let $m \geq 5$ be odd, $T_m = \sum_{i=0}^{m-1} X^{2^i}$, and let $d \in \mathbb{F}_q^*$ satisfy $\mathrm{Tr}(d) = \mathrm{Tr}(d^{-1}) = 0$. Put $A_d = T_m + dX^{2^{m-2}}$ and let p_d be the reduced polynomial of the function $x \mapsto A_d(x)^3$. Then p_d satisfies (H) with $\rho = d^{-4}$, $\mu = 0$ and $Z = \emptyset$. The number of such d is $(q - 3 + \mathcal{K}_m)/4$ with $\mathcal{K}_m = \sum_{x \in \mathbb{F}_q^*} (-1)^{\mathrm{Tr}(x + x^{-1})}$; this number is positive for every odd $m \geq 5$.*

Proof. Write $A = \sum_i a_i X^{2^i}$ for an invertible $\mathbb{F}_2$-linear map, indices modulo m. As a function $A(x)^3 = A(x)^2 A(x) = \sum_{i,j} a_i^2 a_j x^{2^{i+1}+2^j}$, and the reduced polynomial has the Dembowski–Ostrom

coefficients $c_{uv} = a_{u-1}^2 a_v + a_{v-1}^2 a_u$ at $X^{2^u+2^v}$ $(u \neq v)$ and the linear coefficients $a_{w-2}^2 a_{w-1}$ at X^{2^w}. Only X^{1+2^v} and X have odd exponent, so $\varepsilon = a_{m-2}^2 a_{m-1}$ and $K = \sum_{v\geq 1} c_{0v} X^{2^v}$, which as a function equals $a_{m-1}^2 A + a_0 A^2 = a_0 A(A+r)$ with $r = a_{m-1}^2/a_0$ when $a_0 a_{m-1} \neq 0$ (as below). Hence $\ker K = A^{-1}\{0, r\}$, so $\rho = A^{-1}(r)$, and $K(\mathbb{F}_q) = a_0 r^2\{v^2+v\} = a_0 r^2 \ker \mathrm{Tr}$. In $p(X+\rho)+p(X)$ only the terms $c_{1v}X^{2+2^v}$ contribute to X^2, which gives $\mu = \sum_{v\neq 1} c_{1v}\rho^{2^v} = a_0^2 r + a_1 r^2$. With $w = A(a)$ we have $p(a+\rho)+p(a) = (w+r)^3+w^3$ and $p'(a) = \varepsilon + a_0(w^2+rw)$, so $\Lambda(a) = (r+a_0\rho)(w^2+rw)+r^3+\rho\varepsilon$.

For $A = A_d$, $m \geq 5$ gives $a_0 = a_1 = a_{m-1} = 1$ and $a_{m-2} = 1+d$, so $r = 1$ and $\mu = 1+1 = 0$. If $A_d(x) = 0$ then $dx^{2^{m-2}} = \mathrm{Tr}(x)$; $\mathrm{Tr}(x) = 0$ forces $x = 0$, and $\mathrm{Tr}(x) = 1$ forces $x = d^{-4}$ and $\mathrm{Tr}(d^{-1}) = 1$. So A_d is invertible, p_d is a reduced quadratic APN permutation (the cube is one for odd m), and $A_d(d^{-4}) = \mathrm{Tr}(d^{-1}) + 1 = 1 = r$ gives $\rho = d^{-4}$. Next $\varepsilon = (1+d)^2$ and $K(\mathbb{F}_q) = \ker \mathrm{Tr}$, so $\varepsilon \notin K(\mathbb{F}_q)$ because $\mathrm{Tr}(1+d) = 1$; thus (H) holds. Finally $r + \rho = 1 + d^{-4} \neq 0$ since $d \neq 1$, and

$$\frac{r^3+\rho\varepsilon}{(r+\rho)r^2} = \frac{d^4+d^2+1}{d^4+1} = 1 + \Big(\frac{1}{d+1} + \frac{1}{(d+1)^2}\Big)^2$$

has trace 1, so the constant of the affine map Λ is not in the image of its linear part, $(1+\rho)\ker \mathrm{Tr}$, and $Z = \emptyset$. For the count, $\frac{1}{4}\sum_{d\neq 0}(1+(-1)^{\mathrm{Tr}d})(1+(-1)^{\mathrm{Tr}d^{-1}}) = \frac{1}{4}(q-3+\mathcal{K}_m)$, and the Weil bound $|\mathcal{K}_m| \leq 2\sqrt{q}$ for Kloosterman sums makes this positive for $q \geq 32$. □

Within the family the conditions are also necessary: A_d is singular when $\mathrm{Tr}(d^{-1}) = 1$, and $\varepsilon \in K(\mathbb{F}_q)$ when $\mathrm{Tr}(d) = 1$. They coincide with the admissibility conditions of the trace family below. If all coefficients of p lie in $\mathbb{F}_2$, then $\rho = 1$, $\varepsilon = 1$ and $p(0)+p(1) = 1$ because p permutes $\mathbb{F}_2$, so $\Lambda(0) = 0$ and $Z \neq \emptyset$; representatives with $\mu = 0$ and $Z = \emptyset$ therefore need coefficients outside $\mathbb{F}_2$. The number of admissible d is $10, 28, 126, 528, 2002$ for $m = 5, \ldots, 13$. The field-level part of the proof (invertibility of A_d, $A_d(d^{-4}) = 1$, $\ker K = \{0, d^{-4}\}$, $\varepsilon \notin K(\mathbb{F}_q)$ and the absence of zeros of Λ) is formalized (Appendix A); the coefficient formulas for p_d and the count are on paper. We checked Theorem 10.1 for every admissible d with $m \leq 13$ by evaluating p_d at all points (including a full difference table for $m \leq 9$), for sampled d up to $m = 31$ symbolically, and for several d with the independent invariant program of Appendix A. Combined with Theorem B, every odd $m \geq 5$ thus has lifts with $\Delta_{\mathrm{exc}} = q$ at every level $k \geq 3$. The representatives p_d are not sparse: p_d has $3m-3$ monomials. A second family has a bounded number of monomials.

Theorem 10.2 (A sparse family)**.** *Let $m \geq 5$ be odd, $c \in \mathbb{F}_q \setminus \mathbb{F}_2$, $B_c = X^{q/4}+X^{q/2}+X+X^2+cX^4$, and let p_c be the reduced polynomial of $x \mapsto B_c(x)^3$; it has 16 monomials for $m \geq 7$, 12 for $m = 5$, and degree $3q/4$. If B_c is injective, then p_c satisfies (H) with $\varepsilon = 1$, $\rho = B_c^{-1}(1)$, $\mu = 0$ and $Z = \emptyset$. For every odd $m \geq 5$ at least two such c exist; there are $15, 42, 126, 506, 2496$ of them for $m = 5, 7, 9, 11, 13$.*

Proof. In the proof of Theorem 10.1 take $a_{m-2} = a_{m-1} = a_0 = a_1 = 1$, $a_2 = c$ and $a_i = 0$ otherwise. Then $r = 1$, $\varepsilon = 1 \notin K(\mathbb{F}_q) = \ker \mathrm{Tr}$, and $\mu = a_0^2 r + a_1 r^2 = 0$, so (H) holds. Since $B_c(1) = c \neq 1$ we have $\rho \neq 1$, and $\Lambda(a) = (1+\rho)(w^2+w)+1+\rho = (1+\rho)(w^2+w+1)$ with $w = B_c(a)$; for odd m the polynomial w^2+w+1 has no root in $\mathbb{F}_q$, so $Z = \emptyset$.

For existence, write $x \neq 0$ as $x = y^4$ and put $z = y^{-1}$. Then $B_c(x) = y + y^2 + y^4 + y^8 + cy^{16}$ vanishes exactly when $c = h(z)$ with $h = Z^{15}+Z^{14}+Z^{12}+Z^8$. So every $x \neq 0$ lies in the kernel of exactly one B_c, and counting $\sum_c(|\ker B_c| - 1) = q-1$ gives $\#\{c : B_c \text{ injective}\} = 1 + \sum_{\ker B_c \neq 0}(|\ker B_c| - 2)$. This is at least 3 as soon as some kernel has dimension at least 2, that is, as soon as $h(z) = h(w)$ for some $z \neq w$ in $\mathbb{F}_q^*$. Since B_0 is singular and $\mathbb{F}_2$ contains only one further value, at least two c then remain. The pairs with $h(z) = h(w)$, $z \neq w$, are the points of the plane curve $Q(z,w) = (h(z)+h(w))/(z+w) = 0$ of degree 14; neither $z = w$ (where $Q = z^{14}$)

nor the axes are components. Over $\mathbb{F}_{2^{17}}$ this curve has 125769 affine points. If no absolutely irreducible component of Q were defined over $\mathbb{F}_2$, the Frobenius orbits of the components would have lengths between 2 and 14, none dividing the prime 17, so every $\mathbb{F}_{2^{17}}$-point would lie on two distinct components, namely on a component Y and on its conjugate $\sigma^{17}(Y) \neq Y$, and by Bézout there would be at most $14^2/2 = 98$ such points. Hence Q has an absolutely irreducible component Y of degree $e \leq 14$ defined over $\mathbb{F}_2$. By the Weil bound for singular curves of Aubry and Perret [2], Y has at least $q+1-(e-1)(e-2)\sqrt{q}$ points over $\mathbb{F}_q$ on its projective closure; removing at most e points at infinity and at most $3e$ points on the lines $z=0$, $w=0$, $z=w$ leaves a positive number for $q \geq 2^{15}$. The values for $m \leq 13$ were computed exhaustively. □

The counts for $m \leq 13$ and the point count over $\mathbb{F}_{2^{17}}$ were each obtained by two independent programs. We checked Theorem 10.2 symbolically for every admissible c with $m \leq 13$ and for samples up to $m = 31$.

Optimal lifts at no extra cost. For a representative with $\mu = 0$, optimality at every level only asks that A_F have no zero (Theorems C and 6.3), and one coefficient digit suffices.

Proposition 10.3. *Let p satisfy (H) with $\mu = 0$, and let F be a lift of p whose exponents have binary weight at most 2, for instance any lift with the monomials of p. Then $A_F = A_F(0) + L_p$ with an $\mathbb{F}_2$-linear map L_p that does not depend on F and satisfies $L_p(\rho) = 0$. Let $j \geq 1$ be such that X^{1+2^j} has a nonzero coefficient in p (such j exists since $K \neq 0$), let $\tilde{b} \in R_k$ be its coefficient in F, with residue b, and let F_u be F with $\tilde{b}$ replaced by $\tilde{b} + 2u$. Then $A_{F_u} = A_F + \overline{u}\rho^{2^j}$, and F_u is optimal at every level $k \geq 3$ for at least half of the residues $\overline{u}$. It has the same monomials and degree as F, and if $b \notin \mathbb{F}_2$ it costs no extra operation in the termwise evaluation model of Section 7.*

Proof. Compute $A_F(a)$, the residue of $(F'(x+\tau) - F'(x))/2$ with $\overline{x} = a$, $\overline{\tau} = \rho$, monomial by monomial; let bX^e be a monomial of F. Exponents $e = 1$ and $e \equiv 0 \pmod 4$ contribute no x-dependent term, $e = 2+2^v$ ($v \geq 2$) the term $\overline{b}(a\rho^{2^v} + \rho a^{2^v})$, and, since $(x+\tau)^{2^v} \equiv x^{2^v} + \tau^{2^v} + 2(x\tau)^{2^{v-1}} \pmod 4$, $e = 1 + 2^v$ ($v \geq 1$) the term $\overline{b}(a\rho)^{2^{v-1}}$. The x-independent parts of all monomials add up to an element of $2R_k$, because the total and the x-dependent parts do, so they contribute a constant. Hence $A_F(a) = A_F(0) + L_p(a)$ with $L_p(a) = K(\sqrt{a\rho}) + \rho K_1(a) + \mu a$, where $c_{uv} = c_{vu}$ is the coefficient of $X^{2^u+2^v}$ in p and $K_1 = \sum_{v \neq 1} c_{1v} X^{2^v}$; the coefficients of p alone determine L_p, and $L_p(\rho) = K(\rho) + \rho K_1(\rho) + \mu\rho = 0$ because $K_1(\rho) = \mu$. So $A_F(\mathbb{F}_q)$ has at most $q/2$ elements. Proposition 4.2(i) with $H = uX^{1+2^j}$ adds the constant $\overline{u}\rho^{2^j}$, and A_{F_u} has no zero if and only if $\overline{u}\rho^{2^j} \notin A_F(\mathbb{F}_q)$. □

For $P_{m,r}$ with $r \geq 5$ we have $\rho = 1$, and the coefficient 1 of $X^{2^{r-2}+1}$ replaced by 3 gives $A = A_{F^\sharp}$: an optimal lift with exactly the monomials of the natural lift, at the cost of two additions and no multiplication in the termwise model, one multiplication less than $F^\sharp$ ($F^\sharp$ is the case $j = 1$, where X^3 is a new monomial). Theorems 8.1, 8.2 and 8.4(ii),(iii) depend on the lift only through A_F (given c_3, π and μ), so the proof of Corollary 8.5 applies verbatim and this lift is also optimal in every layer below the top. For $r \geq 5$ it therefore improves on $F^\sharp$ in cost and gives the same exceptional and layer spectra. For p_c the coefficient $1+c$ of X^5 ($j = 2$) gives optimal lifts with the monomials of p_c (16 for $m \geq 7$, 12 for $m = 5$) and no extra operation; the digit is found by linear algebra. With $f = t^5 + t^2 + 1$, $t^7 + t + 1$ and $t^9 + t^4 + 1$, the natural lift itself is optimal for 13 of the 15 values of c at $m = 5$, 28 of 42 at $m = 7$ and 83 of 126 at $m = 9$; these counts depend on the convention for lifting coefficients. At $m = 5$, $k = 3$ we computed all exceptional rows of the 13 optimal natural lifts and of two digit-changed lifts F_u, and those of a further explicit optimal lift (given in the notes in Online Resource 1) for all 15 values of c, and at $m = 7$, $k = 3$ all 16384 rows

of that lift for two values of c: every exceptional entry is 0 or q, as predicted. Natural lifts and digits $\overline{u}$ with zeros of A give entries q^2, also as predicted.

The trace family of Bartoli and Stănică. Let $m \geq 5$ be odd, let $T_m = \sum_{i=0}^{m-1} X^{2^i}$ be the absolute trace polynomial and let $w \in \mathbb{F}_q^*$ satisfy $\mathrm{Tr}(w) = \mathrm{Tr}(w^{-1}) = 0$. The polynomials $F_w = X^3 + wX^2T_m + w^2XT_m + w^3T_m$ of [3, Thm. 2.4] satisfy (H): they are reduced and quadratic, and $F'_w = w^3 + K$ with $K = (1+w)A_w^2 + w^2A_w$, $A_w = X + wT_m(X)$, $\ker K = \{0, \rho\}$ and $\rho = A_w(w^2/(1+w))$. For $m = 5$ and $w = t$ (degree 18) we find $\rho = 31$, $\mu = 13$ and $Z = \emptyset$. So this member lies in the cell where the optimum is q at $k = 3$ and $2q$ at $k \geq 4$, and no family is optimal at all levels (Corollary 6.4). Indeed its natural lift has A zero-free and exceptional rows "64 appears 512 times" at $k = 3$. It is optimal at every $k \geq 4$ by Theorem C. The lift with $A \equiv 0$ has rows "32 appears 1024 times" at $k = 3$, but its rows reach $q^2 = 1024$ at $k = 4$. For $m = 5, 7, 9$ every admissible w gives $\mu \neq 0$. Half of them have $Z = \emptyset$, and the other half have $|Z| = 2$. The constant-derivative family of [3, Thm. 2.3] lies outside (H), since it has no exceptional direction.

Open problems. (1) Closed formulas for the parameters d and c of Theorems 10.1 and 10.2, and for a good digit $\overline{u}$ in Proposition 10.3, that is, a closed-form optimal lift without new monomials. (2) Closed formulas for the residual configurations of the intermediate layers, and expressions of the level data of T through the coefficients of F. (3) The layer values D_s of Proposition 9.4 for $r = 3$ and for general lifts. (4) Per-key bounds for single differentials beyond the averaging argument of Proposition 9.7.

Statements and Declarations

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Data and code availability. The programs, inputs, logs and result files of the finite computations of Appendix A, the replay scripts `replay_core.sh` and `replay_full.sh` with their negative controls, and the Lean 4 sources with their build, audit and kernel re-check records are provided as Online Resource 1 (archive `ESM_1.zip`). The core replay takes about six minutes on two cores and needs gcc and Python 3 with SymPy and NumPy; the full replay takes about one and a half hours. The same material, with the blueprint of the formalization, is archived at `https://doi.org/10.5281/zenodo.23683493` (source repository `https://github.com/hirokifukui/residue-apn`, tag `v2.0.0`).

Author contribution. The author is the sole author of this work and carried out all parts of it.

Use of generative AI. AI tools were used in this work as described in Appendix A.

A Verification

Use of AI tools. Claude (Anthropic), including Claude Code, and ChatGPT (OpenAI) were used to explore and draft proofs, to write the Lean code and the programs for the finite computations,

to draft and revise the text, and to review successive drafts. The model versions changed during the project; representative prompts are available on request. The author made all mathematical and editorial decisions, checked every definition, statement, proof and citation, and takes sole responsibility for the content. The correctness of the formalized results rests on the Lean kernel, and the reported spectra were cross-checked by independently written programs; neither check depends on accepting an assertion made by an AI tool.

Computations. Exceptional rows were counted by a C program over $\mathrm{GR}(2^k, m)$ and compared with the predictions of Theorems 5.1 and 5.2 by a fail-closed validator. The validator rejects empty or unsupported scopes and pins the expected record counts. Rows outside the scope of the spectrum theorems are checked only against the proven lower bound q^2 and are not counted as verified spectra. For $m = 5$ all 1024 exceptional rows at $k = 3$ and sampled rows at $k = 4$ were checked for 16 lifts at $k = 3$ and $k = 4$, giving 32 records: 27 spectrum matches and 5 records outside the spectrum theorems that were checked against the lower bound q^2. For $m = 7$ all 16384 exceptional rows were checked for the representative above, for $F^\sharp$ and the natural lift of $P_{7,3}$, and for $F^\sharp$ and the natural lift of $P_{7,5}$ (whose rows reach exactly q^2, with $n_{2q} = 8064$ and $n_{q^2} = 2$ as in Corollary 7.5). For $m = 9$ two sampled exceptional rows each of the natural lifts of $P_{9,5}$ and $P_{9,7}$ over $\mathrm{GR}(8, 9)$ were computed in full (2^{27} inputs per row); they are "2^{10} appears 130560 times, 2^{18} twice" and "2^{10} appears 130560 times, 2^{19} once", as predicted. An independent Magma implementation agrees on 64 distinct directions for four lifts. The two-word normalization of Proposition 9.3 was checked exhaustively on a two-word model over $\mathbb{Z}/8$ (258048 characteristics), and the bound of Proposition 9.6 exhaustively on two words over $\mathrm{GR}(4, 2)$ for 32 random valuation-preserving permutations (all input and output differences, exact integers). The replay scripts are fail-fast; they reject empty, unknown or repeated stage lists, report a partial run as partial, and are tested by negative controls.

Lean. The central results are formalized in Lean 4 (Mathlib) over an abstract axiom system for Galois rings, and $\mathrm{GR}(2^k, m) = (\mathbb{Z}/2^k)[X]/(f)$ is shown to satisfy it. The formalization covers the invariants and their control, the lower bounds, the minimax theorem and all minimizers ($k = 3$ and $k \geq 4$), the spectra, all layer results (Theorems 8.1, 8.2, 8.4 with the multiplicities of (ii), Corollaries 8.3 and 8.5), the residue APN and permutation properties of $P_{m,r}$, Theorem A on all unit directions together with its degree statement, the natural-lift benchmarks of Theorem A ($\Delta_{\mathrm{exc}} = 4q$ for $r = 3$ at every level, an entry $\geq q^2$ in every exceptional row for $r \geq 5$ at every level, and the full spectrum of Corollary 7.5), the literal comparison of difference tables across rings in Theorem 5.2, Proposition 9.3 for an arbitrary finite additive commutative group, S-box and additive automorphism, part (a) of Corollary G (for any lift whose unit rows are bounded by D, any M whose reduction has branch number B, and any characteristic with a unit input component, the probability is at most $(D/|R_k|)^{B\lfloor N/2 \rfloor}$; for $F^\sharp$ with $D = 2q$ and no hypotheses beyond those of the corollary), and the congruence statement of Proposition 9.8 for any commutative ring, polynomial S-box, additive layer and key schedule (the comparison with a random permutation is on paper), the field-level part of Theorem 10.1, Proposition 9.4 (for any lift whose derivative has no zero and any bound D_s, the top-layer value for every such lift, and $F^\sharp$ with $r \geq 5$ in every layer), Proposition 9.5 for two rounds over any finite abelian group of words, the transfer of the branch number of $\overline{M}$ to R_k^n, Proposition 9.6 for two rounds (for any polynomial S-box that permutes R_k, reduces to a permutation and has p' zero-free, in every layer s and with any bound D_s), and Proposition 9.7; the instances $\eta = 2/q$, $D_0 = 2q$ and $D_s = q^{s+1}$ ($1 \leq s \leq k-2$) are formalized for $F^\sharp$ with $r \geq 5$, and the top-layer instance $D_{k-1} = 2q^{k-1}$ for every lift whose p' is two-to-one.

Optimality across levels is formalized in two forms. Theorem 6.3 and the existence statement for one fixed pair of levels are formalized abstractly and on $\mathrm{GR}(2^k, m) \to \mathrm{GR}(8, m)$. Corollary 6.4, with its quantifier over all levels, is formalized for an abstract tower of Galois rings over a common coefficient ring, and this tower is constructed for the concrete rings: for a monic $g \in \mathbb{Z}[X]$ with irreducible reduction modulo 2, the rings $\mathrm{GR}(2^{j+3}, m) = (\mathbb{Z}/2^{j+3})[X]/(g)$, $j \geq 0$, with coefficients taken from $\mathbb{Z}[X]/(g)$, satisfy its hypotheses, so Corollary 6.4 holds formally on $\mathrm{GR}(2^k, m)$ for all levels at once. All audited declarations depend only on the standard axioms, and the compiled modules were re-checked by the kernel replay tool `leanchecker`. Not formalized: the operation count of Theorem A, the passage from two to $N \geq 2$ rounds in Propositions 9.5 and 9.6, the value $\eta = 2/q$ for a general lift, the statement of Theorem A on which monomials occur (its degree statement is formalized), the existence part and the coefficient formulas of Theorem 10.2 (its finite certificate, the irreducibility of $t^{17} + t^3 + 1$, the 125769 points and the counts for $m \leq 13$, is recomputed by a separate program in the replay), Proposition 10.3 for a general p (for $P_{m,r}$ the equality of A_F with that of $F^\sharp$ and the resulting unit-direction bound are formalized; its layer optimality is on paper), Proposition 8.6 and the classification of the intermediate layers built on it, and the finite computations. A theorem-level scope table is part of Online Resource 1.

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